# MedOrchestrate: Evidence-Grounded Multimodal Clinical Decision Support with Human Review

Project report submitted in partial fulfilment of the requirements for the award of the degree of

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in

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Submitted by


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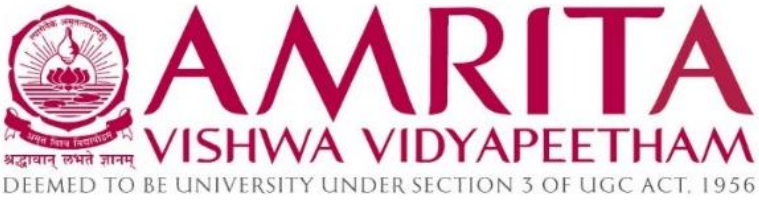
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**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**

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# BONAFIDE CERTIFICATE

This is to certify that the project report entitled "MedOrchestrate: Evidence-Grounded Multimodal Clinical Decision Support with Human Review", submitted by [Student 1 name] ([Student 1 registration number]), [Student 2 name] ([Student 2 registration number]), [Student 3 name] ([Student 3 registration number]) and [Student 4 name] ([Student 4 registration number]), in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering at Amrita Vishwa Vidyapeetham, is a bonafide record of the work carried out by them under my guidance and supervision at Amrita School of Computing, Amritapuri, during Semester 7 of the academic year 2026 - 2027.

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# DECLARATION

We, [Student 1 name] ([Student 1 registration number]), [Student 2 name] ([Student 2 registration number]), [Student 3 name] ([Student 3 registration number]) and [Student 4 name] ([Student 4 registration number]), hereby declare that the project entitled "MedOrchestrate: Evidence-Grounded Multimodal Clinical Decision Support with Human Review" is a record of our original work carried out under the guidance of [Guide name and designation], Department of Computer Science and Engineering, Amrita Vishwa Vidyapeetham, Amritapuri. To the best of our knowledge, this work has not formed the basis for the award of any degree, diploma, fellowship or similar qualification to any candidate at any university. All sources and contributions of others have been appropriately acknowledged.

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This work rests on openly available resources: the U.S. National Library of Medicine (MedlinePlus and PubMed), Europe PMC and ClinicalTrials.gov, whose public interfaces provide the evidence the system retrieves; the authors of BiomedCLIP, BioBERT and MedCPT, who released their models; and the photographers and authors of the openly licensed medical images listed in Appendix A.

# ABSTRACT

Language-model assistants can sound authoritative while being unsupported, and a clinician needs to know what a suggestion rests on and when not to trust it. This project builds MedOrchestrate, a clinician-facing prototype that accepts free text, a medical report, laboratory values, history and an optional CT, MRI or chest X-ray image, and returns a ranked differential in which every diagnosis cites its evidence. The system retrieves from a curated 67-passage knowledge base (BioBERT dense retrieval, MedCPT reranking), queries four public literature sources, reads images with BiomedCLIP zero-shot classification, and fuses everything in a multimodal language model. A deterministic confidence score, not the model's own, decides whether the system refines its query (at most three rounds) and whether a clinician must review the case; clinician corrections are stored as retrievable precedent.

**Achieved:** a working end-to-end system with 641 backend and 124 frontend automated tests; a retrieval harness in which mean reciprocal rank on 14 real agent queries rose from 0.685 to 0.863 through fixes and then fell to 0.764 (hit@5 still 14/14) when the knowledge base grew from 42 to 67 passages to cover missing conditions; an 88-image real-image benchmark on which zero-shot top-1 finding accuracy was 42% on the first 62 (chance 10%), and 56% after a prompt change whose held-out gain (7/25 to 12/25 images) was not significant and was not reproduced on 26 fresh images (15/26 against 14/26); and a live 31-case run that exposed, and led to the fix of, a flaw letting empty inputs skip review. A re-run of all 31 cases on the final system confirmed the fix and a new cap for general-symptom input: all six abstention cases went to review (one before), and review rose from 3 to 13 of 31. An LLM judge (the same model family as the system) marked 28 of 31 differentials correct against a developer-drafted key (30 of 31 in a later run). A second run of the 31 cases moved confidence by 0.02 on average and the review decision for 4 of 31 cases, all near the threshold. A weak-image failure found by the judge was fixed, and the multimodal model's own reading of the raw image was measured (68 of 88 against BiomedCLIP's 50). A cross-check of the zero-shot label against that reading marked three doubtful image reads in 18 image cases as contested without changing any review decision. **Not yet done:** an independent judge, calibration of the confidence threshold, and clinician evaluation (3 real reviews of the 10 required). The project addresses UN Sustainable Development Goal 3, Good Health and Well-being (target 3.8); its contribution is expected, not demonstrated.

**Keywords:** clinical decision support; retrieval-augmented generation; multimodal fusion; confidence calibration; human-in-the-loop

# TABLE OF CONTENTS

**ACKNOWLEDGEMENT** ..... **i**
**ABSTRACT** ..... **ii**
**Chapter 1 Introduction** ..... **1**
1.1 Background and motivation ..... 1
1.2 Objectives ..... 2
1.3 Scope and constraints ..... 3
1.4 Project report organization ..... 3
**Chapter 2 Problem Definition** ..... **4**
2.1 Problem statement ..... 4
2.2 Existing system and limitations ..... 4
**Chapter 3 Literature Review** ..... **6**
3.1 Review scope and selection ..... 6
3.2 Descriptive review of literature works ..... 6
3.3 Comparison of reviewed literature ..... 12
**Chapter 4 Requirements** ..... **13**
4.1 Functional requirements ..... 13
4.2 Non-functional requirements ..... 14
4.3 Hardware requirements ..... 14
4.4 Software and data requirements ..... 14
**Chapter 5 Proposed Methodology** ..... **16**
5.1 Design (architecture) or conceptual framework ..... 16
5.2 Proposed approach and procedure ..... 17
**Chapter 6 Results and Analysis** ..... **23**
6.1 Evaluation setup ..... 23
6.2 Evaluation criteria ..... 23
6.3 Result ..... 24
6.4 Comparative analysis ..... 37
6.5 Discussion and Interpretation ..... 37
6.6 Limitations ..... 38
**Chapter 7 Conclusion** ..... **39**
7.1 Conclusion ..... 39
7.2 Future work ..... 39
**References** ..... **41**
**Appendix A Source Code and Reproducibility** ..... **42**
A.1 Setup and execution ..... 42
A.2 Supporting material ..... 42
**Appendix B Supporting Results** ..... **43**
B.1 Per-query retrieval results ..... 43
B.2 Per-finding imaging results ..... 43
**Appendix C Worked Example** ..... **45**
C.1 Input ..... 45
C.2 Output ..... 45

**Appendix D Knowledge Base Inventory ..... 46**
**Appendix E API and Configuration Reference ..... 48**
E.1 HTTP interface ..... 48
E.2 Configuration ..... 48
**Appendix F Test Inventory and Live-Case Results ..... 49**
F.1 Automated tests by module ..... 49
F.2 The 31 live cases ..... 50

# LIST OF FIGURES

**Figure 5.1: System architecture ..... 16**
**Figure 5.2: Pipeline workflow with the confidence-driven refinement loop and clinician review ..... 17**
**Figure 5.3: Two-stage retrieval and calibrated scoring ..... 19**
**Figure 5.4: Asynchronous diagnosis: submit, progress events and polling..... 22**
**Figure 5.5: The interface: (a) case form with history and lab fields; (b)–(d) screens populated by a mock API, so their contents are illustrative and not system output... 22**
**Figure 6.1: Retrieval quality as the pipeline was corrected, and after the knowledge base grew ..... 25**
**Figure 6.2: Top-1 finding accuracy before and after the caption change, with Wilson intervals ..... 27**
**Figure 6.3: Prompt-wording experiment: development choice and the one-time held-out result..... 28**
**Figure 6.4: Reported confidence for the 31 live cases: first run (before the input-sufficiency gate, grey) and the run on the final system (coloured) ..... 32**

# LIST OF TABLES

**Table 1.1: Objectives, evaluation methods, deliverables and status ..... 2**
**Table 2.1: Existing approaches and the gap this project addresses ..... 4**
**Table 3.1: Comparison of reviewed literature ..... 12**
**Table 4.1: Functional requirements and acceptance criteria ..... 13**
**Table 4.2: Non-functional requirements ..... 14**
**Table 4.3: Software and data used ..... 14**
**Table 5.1: Composition of the knowledge base ..... 18**
**Table 5.2: Parameters of the confidence score ..... 20**
**Table 6.1: Retrieval on 14 real agent queries (hit@5, hit@1 and MRR@5); 42-passage knowledge base unless stated ..... 24**
**Table 6.2: Zero-shot finding accuracy on real images (true modality given; Wilson 95% interval) ..... 26**
**Table 6.3: The 17 fresh images under the current captions and the bare labels ..... 28**
**Table 6.4: Judge results over the 31 final reports (a consistency check, not accuracy) ..... 31**
**Table 6.5: Raw-image reading by the multimodal model against BiomedCLIP, same 88 images ..... 34**
**Table 6.6: Zero-shot reads contested by the second reading, with the cross-check off and on ..... 35**
**Table 6.7: Results against baselines and expectations ..... 37**
**Table 7.1: Next steps, dependencies and proposed validation ..... 39**
**Table B.1: Retrieval harness, final system: rank of the first relevant passage ..... 43**
**Table B.2: Top-1 per finding on the original 62 images, before and after the caption change (counts of images) ..... 43**
**Table C.1: Case CASE002 as submitted ..... 45**
**Table C.2: Ranked differential for CASE002 ..... 45**
**Table D.1: Knowledge-base passages ..... 46**
**Table E.1: Endpoints ..... 48**
**Table E.2: Settings read from the environment (documented in .env.example) ..... 48**
**Table F.1: Backend tests collected by pytest ..... 49**
**Table F.2: Per-case results: first run (before the input-sufficiency gate) and the run on the final system ..... 50**

# Chapter 1
# Introduction

## 1.1 Background and motivation

Computerised clinical decision support systems help clinicians with complex decisions and have evolved from the 1980s to today's record-integrated tools, but their effects on providers, patient outcomes and costs are not fully known and past setbacks show that they carry real risks [1]. A specific risk is automation bias, the tendency to over-rely on automated advice. A systematic review of 74 studies found that trust, confidence and workload influence it, and that design choices such as providing information rather than recommendations, and showing updated confidence levels, can mitigate it [2]. Any system that suggests diagnoses must therefore show what its suggestion rests on and how far to trust it.

Large language models now encode considerable clinical knowledge, yet human evaluation of a leading model still revealed gaps and answers that remained inferior to clinicians' [5]. Retrieval-augmented generation conditions generation on retrieved text; its authors present this as a way to address the lack of provenance and the difficulty of updating knowledge in purely parametric models [3], and a large benchmark for medicine reports accuracy gains of up to 18% from retrieval over chain-of-thought prompting across six models [4]. Clinical reasoning is also multimodal. BiomedCLIP, a biomedical vision-language model pretrained on 15 million figure-caption pairs [6], scores images against text, so an image can be classified with text labels alone; how reliable that is on real films is an open question that this project measures rather than assumes.

MedOrchestrate is a prototype that combines these ideas under one rule: the language model may reason, but it may not set its own confidence, and a clinician stays in the loop. Evidence is retrieved from a curated knowledge base and from four public literature sources; every listed diagnosis must cite what supports it; a deterministic score, computed from the

evidence and not from the model's self-reported certainty, decides when the system should refine its search and when a human must review the case.

### 1.1.1 Sustainable Development Goal mapping

The project is relevant to **United Nations Sustainable Development Goal 3: Good Health and Well-being** ("Ensure healthy lives and promote well-being for all at all ages") [16], specifically **target 3.8**, universal health coverage including access to quality essential health-care services. The mapping runs through the problem (reliable second-opinion support is unevenly available), the objectives (evidence that is traceable, a confidence that is honest about thin input, and mandatory review), the contribution (a locally runnable prototype that uses only public evidence sources and open models) and the intended beneficiaries (clinicians working with limited specialist access, and their patients). This is an **expected** contribution. Nothing in this report demonstrates a clinical outcome, and the system is not validated for clinical use.

## 1.2 Objectives

The objectives are stated so that each can be checked, and Table 1.1 links each to an evaluation method, a deliverable and its status.

Table 1.1: Objectives, evaluation methods, deliverables and status

| ID | Objective | Evaluation method | Deliverable | Status |
|---|---|---|---|---|
| O1 | Accept any combination of text, report, labs, history and image, and return a ranked differential with cited evidence. | Automated tests (641 backend, 124 frontend); live runs | FastAPI backend, React interface | Met; see Section 6.3.1 |
| O2 | Ground the differential in a curated knowledge base and four live sources; the relevant passage should reach the top five. | 14-query retrieval harness: hit@5, MRR@5 | 67-passage knowledge base; retrieval pipeline | Met on the harness; Section 6.3.2 |
| O3 | Read CT, MRI and chest X-ray images with BiomedCLIP and characterise its reliability with uncertainty. | 88 real images; Wilson intervals; held-out test; fresh-image checks | Image benchmark and report | Met; Section 6.3.3 |
| O4 | Make confidence deterministic, and route thin, empty or low-confidence cases to review. | Unit tests; 31-case live run; abstention check | Confidence module with sufficiency gate | Met mechanically (gate and cap verified by tests and on all 31 live cases); calibration not done; Section 6.3.6 |

| ID | Objective | Evaluation method | Deliverable | Status |
|---|---|---|---|---|
| O5 | Refine at most three times, then hand over to a clinician; keep corrections as retrievable precedent. | Tests; live runs; case-memory round trip | Optimizer, review screen, case memory | Met mechanically; clinical value not evaluated |
| O6 | Show sources, confidence and live progress in the interface. | Browser verification; real async run | Four-screen UI, async progress API | Met; Section 6.3.15 |
| O7 | Report results honestly, separating achieved from planned work. | This report | Chapter 6 and Appendix B | Planned items listed in Section 6.6.1 |

## 1.3 Scope and constraints

**Intended users** are clinicians reviewing a case; the output is decision support for review, never an autonomous diagnosis. **Included:** text, report, laboratory value and history input; CT, MRI and chest X-ray images; offline and live evidence retrieval; multimodal fusion; deterministic confidence with an input-sufficiency gate; a bounded refinement loop; clinician accept, override and annotate; persistence; case memory; weight recalibration; and a web interface with live progress. **Excluded:** DICOM handling, integration with health records, regulatory validation, patient-identity management, and any clinical trial.

**Assumptions and constraints.** The prototype assumes that the four public evidence sources and a language-model provider are reachable. Live runs used a free-tier Gemini model (`gemini-3.5-flash-lite`), whose request quotas repeatedly limited the evaluation; the code supports Claude, but that path was unit-tested and not run end to end. The knowledge base is deliberately small (67 passages). The test cases are 24 synthetic and 7 real-image cases. The expected-diagnosis key is developer-drafted and not clinician-verified. The work was carried out in Semester 7.

## 1.4 Project report organization

Chapter 2 defines the problem and the limits of existing approaches. Chapter 3 reviews fourteen works. Chapter 4 states the requirements. Chapter 5 presents the design and methodology. Chapter 6 reports the evaluation, separating achieved results from planned ones. Chapter 7 concludes and lists future work. Appendix A gives the repository, setup and image credits, and Appendix B gives supporting results.

# Chapter 2
# Problem Definition

## 2.1 Problem statement

**Problem.** Given a patient's age and sex and any combination of free text, a medical report, laboratory values, past history and a CT, MRI or chest X-ray image, produce a ranked differential diagnosis, with the evidence behind each entry, and an honest indication of whether the result can be relied on. **Stakeholders** are the reviewing clinician, who needs traceable evidence; the patient, who is affected by an over-confident suggestion; and the institution, which needs the system to fail visibly rather than silently.

**Consequences of getting it wrong.** An unsupported but fluent answer invites over-reliance [2]; an empty or vague input that still yields a confident diagnosis is worse. The project therefore poses three questions. **RQ1:** can retrieval from a curated knowledge base and live sources make each diagnosis traceable to evidence? **RQ2:** can a deterministic confidence and an input-sufficiency check route unreliable cases to a human? **RQ3:** how reliable is zero-shot imaging as a pre-filter on real images? The inputs are the case fields above; the output is a report with a ranked differential, cited evidence, flagged conflicts, imaging findings, a confidence score and a review flag.

## 2.2 Existing system and limitations

Table 2.1 summarises the families of existing approaches, with their limitations as reported in the literature reviewed in Chapter 3.

Table 2.1: Existing approaches and the gap this project addresses

| Approach | Strength | Limitation (source) | Gap |
|---|---|---|---|
| Clinical decision support in records [1] | Integrated into clinical workflow | Effects on outcomes and costs not fully known; documented risks and pitfalls [1] | Evidence behind each suggestion is rarely shown |
| General medical language models [5] | State-of-the-art accuracy on medical exam benchmarks | Gaps found by human evaluation; answers inferior to clinicians [5] | No provenance or deterministic reliability signal |
| Retrieval-augmented medicine [4] | Accuracy gains of up to 18% over chain-of-thought across six models | No agreed best settings for different medical purposes [4] | Text only; no confidence-driven escalation to a human |

| Approach | Strength | Limitation (source) | Gap |
|---|---|---|---|
| Biomedical vision-language models [6, 7] | Image-text retrieval, classification and question answering; a conversational image assistant | Reliability on real clinical films not reported in the abstracts | Image findings unvalidated; need a conflict check and cautious use |

Among the works reviewed, none combines multimodal evidence, traceable citations, a reliability score that the language model cannot set, and a mandatory human review step in one pipeline; that combination is the gap this project addresses. The statement is limited to the fourteen works in Chapter 3 and is not a claim about the whole field.

# Chapter 3
# Literature Review

## 3.1 Review scope and selection

The review covers the design questions this project had to answer: how decision support fails when advice is trusted without evidence [1, 2]; how language models and retrieval can be combined for medicine [3–5, 9]; which biomedical image-text and retrieval models to build on [6–8]; how to fuse rankings [11] and calibrate a score [10]; and how to evaluate generated answers [12]. Searches used these topics as keywords (clinical decision support, automation bias, retrieval-augmented generation, medical question answering, biomedical vision-language model, biomedical retrieval, rank fusion, calibration, LLM-as-a-judge) on arXiv, PubMed, Europe PMC and publisher pages. Works were selected if they were primary papers whose result bore directly on a design choice, and the publication period spans 2009 to 2025. Fourteen works were read; twelve are reviewed individually below, and two (BioBERT [13] and Ragas [14]) are cited where they are used.

> Reading depth and honesty. Bibliographic details and findings were taken from each paper's own abstract or primary page (for [11], the paper itself). We did not reproduce any result. Where an abstract is silent we write "Not reported", as the template requires, and "Not applicable" only where a detail does not apply.

## 3.2 Descriptive review of literature works

### 3.2.1 Clinical decision support systems: benefits, risks and strategies

Sutton et al. (2020) in "An overview of clinical decision support systems: benefits, risks, and strategies for success" [1] investigate how computerised clinical decision support systems (CDSS) help clinicians with complex decisions, and what risks they carry, as such systems are increasingly delivered through electronic medical records. The work proposes an overview of the types of CDSS, use cases with proven efficacy, common pitfalls and potential harms, and closes with evidence-based recommendations for reducing risk in CDSS design, implementation, evaluation and maintenance. The study is situated in the literature on CDSS since the 1980s; datasets, tools and experiments are not applicable. The evaluation is a qualitative synthesis, and no quantitative criteria are reported in the abstract. The authors report that the effects of CDSS on providers, patient outcomes and costs are not fully known, and that past successes and setbacks show these systems carry risks.

Comparisons with other approaches are not applicable. Limitations are not reported in the abstract. This work informs the present project through its warning that decision support is not automatically safe; it leaves open how to show the evidence behind each individual suggestion, which this project addresses.

### 3.2.2 Automation bias in clinical decision support

Goddard, Roudsari and Wyatt (2012) in "Automation bias: a systematic review of frequency, effect mediators, and mitigators" [2] investigate the tendency to over-rely on automated advice, with a focus on clinical decision support. The work is a systematic review: of 13,821 retrieved papers, 74 met the inclusion criteria. The databases and period searched are not reported in the abstract. The authors synthesise the included studies for the frequency of automation bias, the factors that mediate it and the ways to mitigate it. They report that user factors, trust and confidence, and environmental pressures such as workload, task complexity and time constraints influenced automation bias, and that mitigators included training, emphasising user accountability, and design choices such as where advice appears on screen, updated confidence levels, and providing information rather than recommendations. Comparisons with other approaches are not applicable and limitations are not reported in the abstract. This work informs the present project by motivating a display of evidence and confidence, and a mandatory review step, instead of a bare recommendation. It leaves open which confidence signal deserves trust, which is why this project computes confidence deterministically.

### 3.2.3 Retrieval-augmented generation

Lewis et al. (2020) in "Retrieval-augmented generation for knowledge-intensive NLP tasks" [3] investigate how to give generative language models access to external knowledge, since pretrained models store facts in their parameters but cannot readily access or precisely manipulate them, give provenance for their decisions, or update their knowledge. The work proposes retrieval-augmented generation (RAG) models that combine a pretrained sequence-to-sequence model as parametric memory with a dense vector index of Wikipedia, accessed through a pretrained neural retriever, as non-parametric memory, in two formulations: one conditions on the same retrieved passages for the whole output, the other can use different passages per token. The models were fine-tuned and evaluated on a range of knowledge-intensive tasks. The authors report state-of-the-art results on three open-domain question-answering tasks, outperforming parametric sequence-to-sequence models and task-specific retrieve-and-extract architectures, and on generation tasks more specific,

diverse and factual output than a parametric-only baseline. Limitations are not reported in the abstract. This work informs the present project through its core design of retrieving before generating; the gap is that its corpus is general-domain and it has no notion of when retrieval was not good enough.

### 3.2.4 Benchmarking retrieval-augmented generation for medicine

Xiong, Jin, Lu and Zhang (2024) in "Benchmarking retrieval-augmented generation for medicine" [4] investigate hallucination and outdated knowledge in medical question answering, and the lack of best practices for the optimal RAG setting for different medical purposes. The work proposes MIRAGE, a benchmark of 7,663 questions from five medical question-answering datasets, and the MedRAG toolkit. The study ran over 1.8 trillion prompt tokens across 41 combinations of corpora, retrievers and backbone language models. The authors report that MedRAG improves the accuracy of six language models by up to 18% over chain-of-thought prompting, raises GPT-3.5 and Mixtral to GPT-4-level performance, performs best when corpora and retrievers are combined, and shows a log-linear scaling property and "lost-in-the-middle" effects. Limitations are not reported in the abstract. This work informs the present project by supporting retrieval, and the combination of a curated corpus with live sources. It evaluates question answering rather than case-level, multimodal decision support and has no reliability gate, which this project adds.

### 3.2.5 Large language models encode clinical knowledge

Singhal et al. (2023) in "Large language models encode clinical knowledge" [5] investigate how to assess the clinical knowledge of language models, because existing assessments rely mainly on automated evaluation over limited benchmarks. The work proposes MultiMedQA, which combines six existing open question-answering datasets with HealthSearchQA, a new set of medical questions searched online, a human evaluation framework covering factuality, comprehension, reasoning, possible harm and bias, and instruction prompt tuning, which produced Med-PaLM from Flan-PaLM. PaLM (540 billion parameters) and Flan-PaLM were evaluated. The authors report that Flan-PaLM reached state-of-the-art accuracy on every multiple-choice dataset, including 67.6% on MedQA, over 17% above the prior state of the art, that human evaluation revealed key gaps, and that Med-PaLM performed encouragingly but remained inferior to clinicians. The authors state that the gaps expose important limitations of today's models. This work informs the present project: model knowledge alone is not enough, so answers need grounding and review. It offers no citations to evidence and no deterministic reliability signal.

### 3.2.6 BiomedCLIP

Zhang et al. (2025; arXiv 2023) in "A multimodal biomedical foundation model trained from fifteen million image–text pairs" [6], which introduces BiomedCLIP, investigate the need for high-quality paired multimodal data to build a generalist biomedical model. The work proposes PMC-15M, 15 million image-text pairs from 4.4 million scientific articles, which the authors describe as two orders of magnitude larger than datasets such as MIMIC-CXR, and BiomedCLIP, a multimodal foundation model with domain-specific adaptations. The study is situated in retrieval, classification and visual question answering on standard biomedical benchmarks, with ablations. The authors report new state-of-the-art results across standard datasets and better performance than radiology-specific models such as BioViL on RSNA pneumonia detection. The models are released as open access. Limitations are not reported in the abstract. This work informs the present project as its image pre-filter. Because its pretraining data are published articles, this project flags possible overlap with its own image benchmark, and it measures reliability on individual real films, which the abstract does not report.

### 3.2.7 LLaVA-Med

Li et al. (2023) in "LLaVA-Med: Training a large language-and-vision assistant for biomedicine in one day" [7] investigate how to build a conversational assistant for biomedical images, because general-domain vision-language models lack depth on them and prior conversational work focuses on text. The work proposes a model trained on a PubMed Central figure-caption dataset, with open-ended instruction-following data generated from the captions by GPT-4, and a general-domain vision-language model fine-tuned with a curriculum that first aligns biomedical vocabulary and then teaches conversation. Training took under 15 hours on eight A100 GPUs. The model was tested on three standard biomedical visual question-answering datasets. The authors report strong multimodal conversational ability and better results than previous supervised state-of-the-art methods on certain metrics. Limitations are not reported in the abstract. This work shows that image-and-language assistants for biomedicine are feasible, which is relevant to the multimodal fusion step. It provides no evidence citation and no reliability gate, and it is not used in this project.

### 3.2.8 MedCPT

Jin et al. (2023) in "MedCPT: Contrastive pre-trained transformers with large-scale PubMed search logs for zero-shot biomedical information retrieval" [8] investigate biomedical

information retrieval, which mostly relies on lexical matching because semantic retrieval with language-model encoders needs many query-article annotations that are scarce in biomedicine. The work proposes MedCPT, a contrastively pre-trained retriever and re-ranker that are closely integrated, trained on about 255 million user click logs from PubMed. The authors assess six biomedical retrieval tasks and the quality of article and sentence representations. They report state-of-the-art performance on the six tasks, including better results than much larger models such as the GPT-3-sized cpt-text-XL. Limitations are not reported in the abstract. This work informs the present project directly: its cross-encoder is this project's default reranker. The abstract reports benchmark tasks, not symptom-list queries over a 67-passage corpus, so the present project measures it on its own queries (Chapter 6).

### 3.2.9 Self-RAG

Asai et al. (2023) in "Self-RAG: Learning to retrieve, generate, and critique through self-reflection" [9] investigate factual errors that arise when language models rely on parametric knowledge alone, and the cost of standard RAG, which retrieves a fixed number of passages whether or not retrieval is needed or the passages are relevant. The work proposes Self-RAG, in which one language model is trained to retrieve on demand and to generate and reflect on the retrieved passages and its own output using special reflection tokens, which also make the model controllable at inference time. The authors evaluate open-domain question answering, reasoning, fact verification and long-form generation. They report that the 7B and 13B versions outperform state-of-the-art language models and retrieval-augmented models, including ChatGPT and retrieval-augmented Llama2-chat on question answering, reasoning and fact verification, with significant gains in factuality and citation accuracy for long-form generation. Limitations are not reported in the abstract. This work informs the present project's idea of retrieving again when evidence is inadequate; this project uses a deterministic external score and no model training.

### 3.2.10 On calibration of modern neural networks

Guo, Pleiss, Sun and Weinberger (2017) in "On calibration of modern neural networks" [10] investigate confidence calibration, meaning whether predicted probabilities match how often a model is correct, which matters in many applications. The work evaluates post-processing calibration methods on state-of-the-art architectures, using image and document classification datasets, and identifies depth, width, weight decay and batch normalisation as factors that influence calibration. The authors report that modern neural networks are poorly

calibrated, unlike networks of a decade earlier, and that temperature scaling, a single-parameter variant of Platt scaling, is surprisingly effective at calibrating predictions on most datasets. Limitations are not reported in the abstract. This work informs the present project as a caution: scores that look like probabilities usually are not, and need a calibration step. This project rescales its reranker score with two constants, a similar one-parameter-pair idea, but fits them by hand to match average values and does not claim a calibrated probability (Section Section 5.2.4).

### 3.2.11 Reciprocal rank fusion

Cormack, Clarke and Büttcher (2009) in "Reciprocal rank fusion outperforms Condorcet and individual rank learning methods" [11] investigate how to combine the document rankings of several retrieval systems without training examples. The work proposes reciprocal rank fusion (RRF), which scores a document by the sum over rankings of 1/(k + rank), with k = 60 fixed in a pilot study and not altered later. The study ran four pilot experiments, each combining 30 configurations of one search engine over four TREC collections, then four sets of TREC submissions, and built a meta-learner on the LETOR 3 dataset. The authors report that RRF consistently yields better results than any individual system and better results than Condorcet Fuse, that it equalled or bettered CombMNZ, that k = 60 was near-optimal but not critical, and that the meta-learner ranked LETOR 3 better than any previously reported method. Limitations are not reported. This work informs the optional RRF ordering of retrieved passages in this project; its own evidence on this project's data is small and mixed (Chapter 6).

### 3.2.12 LLM-as-a-judge

Zheng et al. (2023) in "Judging LLM-as-a-judge with MT-Bench and Chatbot Arena" [12] investigate how to evaluate chat assistants with strong language models as judges, since existing benchmarks miss human preferences. The work contributes MT-Bench, a multi-turn question set, and Chatbot Arena, a crowdsourced battle platform. The authors report that strong judges such as GPT-4 agree with controlled and crowdsourced human preferences at over 80%, which they note matches human-to-human agreement, and they identify position, verbosity and self-enhancement biases and limited reasoning ability, for which they propose some mitigations. The context is open-ended chat, not medicine. This work informs the present project's LLM-judged evaluation and, through the self-enhancement bias, the caveat that a model grading its own output is a consistency check and not an accuracy measurement (Section Section 6.6.1).

## 3.3 Comparison of reviewed literature

Table 3.1 compares the twelve works on the attributes the template names. “NR” means not reported in the abstract and “NA” not applicable.

Table 3.1: Comparison of reviewed literature

| Work and citation | Problem or objective | Approach and contribution | Study context | Evaluation criteria | Key findings | Limitations and gap |
|---|---|---|---|---|---|---|
| Sutton et al., 2020 [1] | Benefits and risks of clinical decision support | Overview of CDSS types, efficacy, pitfalls, harms; recommendations | CDSS literature since the 1980s | Qualitative synthesis | Effects on outcomes and costs not fully known; real risks | NR. No per-suggestion evidence |
| Goddard et al., 2012 [2] | Over-reliance on automated advice | Systematic review: frequency, mediators, mitigators | 74 of 13,821 papers included | Synthesis of included studies | Trust and workload drive it; information-not-recommendation and confidence displays mitigate | NR. Which confidence to trust is open |
| Lewis et al., 2020 [3] | Provenance and updating of model knowledge | RAG: seq2seq model + dense Wikipedia index | Wikipedia; knowledge-intensive NLP tasks | Open-domain QA, generation | State of the art on three QA tasks; more factual generation | NR. General domain; no reliability notion |
| Xiong et al., 2024 [4] | Hallucination and stale knowledge in medical QA | MIRAGE benchmark and MedRAG toolkit | 7,663 questions, 5 datasets, 41 setups | Accuracy vs chain-of-thought | Up to 18% gain across six LLMs; combine corpora and retrievers | NR. QA only; no case-level or gating |
| Singhal et al., 2023 [5] | Assess clinical knowledge of LLMs | MultiMedQA; human evaluation; instruction prompt tuning (Med-PaLM) | Six QA datasets + HealthSearchQA | Accuracy; human rating of factuality, harm, bias | 67.6% on MedQA; gaps found; below clinicians | Gaps stated. No citations or reliability signal |
| Zhang et al., 2025 [6] | Paired biomedical image-text data and model | PMC-15M and BiomedCLIP | 15M pairs, 4.4M articles | Retrieval, classification, VQA | State of the art; beats BioViL on RSNA pneumonia | NR. Single-film reliability not reported |
| Li et al., 2023 [7] | Conversational assistant for biomedical images | GPT-4-generated instruction data; curriculum fine-tuning | PMC figure-caption data; <15 h on 8 A100 | Three biomedical VQA datasets | Strong conversation; beats prior supervised SOTA on some metrics | NR. No citations or gate |
| Jin et al., 2023 [8] | Semantic biomedical retrieval without annotations | MedCPT retriever and re-ranker | 255M PubMed click logs | Six retrieval tasks; representation quality | State of the art; beats larger cpt-text-XL | NR. Not tested on symptom-list queries |
| Asai et al., 2023 [9] | Fixed-count retrieval; factual errors | Self-RAG with reflection tokens | QA, reasoning, fact verification, long-form | Task accuracy; factuality; citation accuracy | Beats ChatGPT and retrieval-augmented Llama2-chat | NR. Needs training; no external score |
| Guo et al., 2017 [10] | Calibration of predicted probabilities | Evaluate post-processing calibration; temperature scaling | Image and document classification | Calibration of predictions | Modern nets poorly calibrated; temperature scaling effective | NR. Classifiers, not retrieval scores |
| Cormack et al., 2009 [11] | Combine rankings without training data | Reciprocal rank fusion, k = 60 | TREC collections; LETOR 3 | Retrieval effectiveness | Beats any single system and Condorcet Fuse | NR. Not tested on medical retrieval |
| Zheng et al., 2023 [12] | Evaluate chat assistants at scale | LLM-as-a-judge; MT-Bench; Chatbot Arena | Open-ended chat | Agreement with human preferences | >80% agreement; position, verbosity, self-enhancement bias | Not medical; bias when judging own output |

# Chapter 4
# Requirements

## 4.1 Functional requirements

Table 4.1 lists the functional requirements. Each has an acceptance criterion that is checked by an automated test, a benchmark or a live run described in Chapter 6.

Table 4.1: Functional requirements and acceptance criteria

| ID | Requirement | Acceptance criterion |
|---|---|---|
| FR1 | Accept age, sex and any combination of text, medical report, laboratory values, history and an image. | A report is returned for every combination in the 24 synthetic and 7 real-image cases |
| FR2 | Extract symptoms, history, labs and report findings as validated JSON; merge structured history and labs. | Schema-valid output; typed history and labs reach retrieval and fusion |
| FR3 | Detect CT, MRI or chest X-ray and report findings; say “not assessed” for any other modality. | Real-image benchmark; no findings from a wrong label set |
| FR4 | Retrieve and rerank offline evidence and attach clinician precedents. | Relevant passage in the top five on the retrieval harness |
| FR5 | Query four public sources concurrently; isolate failures. | Report flags live evidence unavailable only if all four fail |
| FR6 | Return a ranked differential in which every diagnosis cites existing evidence IDs; flag conflicts; attach the image when supported. | No citation to evidence that was not provided |
| FR7 | Compute confidence deterministically; never use the model’s own certainty. | No diagnosis gives 0; no clinical anchor caps at 0.50 |
| FR8 | Refine below the threshold, at most three rounds. | Never more than three rounds |
| FR9 | Require clinician review when below threshold, gated or degraded; capture accept, override, annotate; store overrides as precedent. | review_required never false for a below-threshold result |
| FR10 | Recalibrate weights only from at least ten distinct reviewed cases, with an audit trail. | No change below ten cases |
| FR11 | Offer an asynchronous endpoint with real step progress, keeping the synchronous endpoint. | Real run: 202 at once, steps move with measured timings |
| FR12 | Provide the four-screen interface with citations, confidence gauge, history and lab entry. | Browser verification |

## 4.2 Non-functional requirements

Table 4.2: Non-functional requirements

| ID | Quality | Requirement and measure |
|---|---|---|
| NFR1 | Safety and transparency | A below-threshold result is never presented as not needing review; failed steps are visible; the confidence is explainable from three signals. |
| NFR2 | Reliability | One failing step or source degrades the run instead of failing it; verified with injected failures and a real provider-quota outage. |
| NFR3 | Privacy and security | No secrets committed; logs redact keys; only synthetic and openly licensed data in the test set. |
| NFR4 | Performance | Single case from about 30 s to several minutes (34 s, 72 s and 505 s observed); the asynchronous endpoint removes the long-held request. |
| NFR5 | Reproducibility | Evaluation harnesses run offline where possible, use fixed seeds and committed inputs, and write their results to files. |
| NFR6 | Usability | No horizontal overflow at a 375 px width; visible keyboard focus; clear error states. |
| NFR7 | Maintainability | Modular agents; 641 backend and 124 frontend tests; documentation of known gaps. |

## 4.3 Hardware requirements

The prototype runs on a single Windows 11 workstation with CPU-only inference; no GPU is required. Several gigabytes of disk hold the Python packages and the model weights (the MedCPT cross-encoder alone is 439 MB, and BiomedCLIP is of similar size). Network access is needed for the four evidence sources and the language-model provider. The processor and memory were not benchmarked, so no minimum specification is claimed.

## 4.4 Software and data requirements

Table 4.3: Software and data used

| Component | Version or source |
|---|---|
| Language and runtime | Python 3.10.11 (development environment); Node.js 22.16 |
| Backend | FastAPI 0.141.1, Uvicorn 0.52.4, Pydantic 2.12.5, CrewAI 1.15.18, SQLite |
| Models and retrieval | sentence-transformers 6.0.0 (BioBERT sentence model, MedCPT cross-encoder), open_clip_torch 3.3.0 (BiomedCLIP), torch 2.13.0 (CPU), Qdrant client 1.19.0 (embedded store) |
| Language-model providers | Gemini through google-genai 2.20.0 (used for every live run, `gemini-3.5-flash-lite`); Anthropic SDK 1.2.0 (implemented, unit-tested) |
| Frontend | React 18, Vite 5, Tailwind CSS 3.4 |
| Containers | Docker Compose: backend (python:3.11-slim, CPU-only torch and torchvision, 1.0 GB image), static frontend and a Qdrant server; downloaded models kept in a volume |

| Component | Version or source |
|---|---|
| Evidence and data | MedlinePlus and PubMed text (knowledge base, 67 verbatim passages); PubMed, MedlinePlus, Europe PMC and ClinicalTrials.gov public APIs (live evidence); Wikimedia Commons images under CC0, CC BY, CC BY-SA or public-domain terms (Appendix A) |
| Test cases | 24 synthetic cases and 7 real-image cases; developer-drafted answer key (not clinician-verified) |

Datasets are partitioned only where an experiment needs it. The imaging benchmark uses a group-level development and test split described in Section 6.3.4; the retrieval harness and live cases use no partition because nothing is trained on them.

# Chapter 5
# Proposed Methodology

## 5.1 Design (architecture) or conceptual framework

Figure 5.1 shows the architecture. A React application talks to a FastAPI backend. Behind the API, an orchestrator built on CrewAI runs one small crew per pipeline step; each crew's only tool wraps ordinary Python code, so the framework sequences the steps while deterministic code does the work. The steps call local models (BiomedCLIP, BioBERT, MedCPT), a language-model provider, an embedded Qdrant store and four public literature sources, and persist cases, reports, reviews and logs in SQLite.

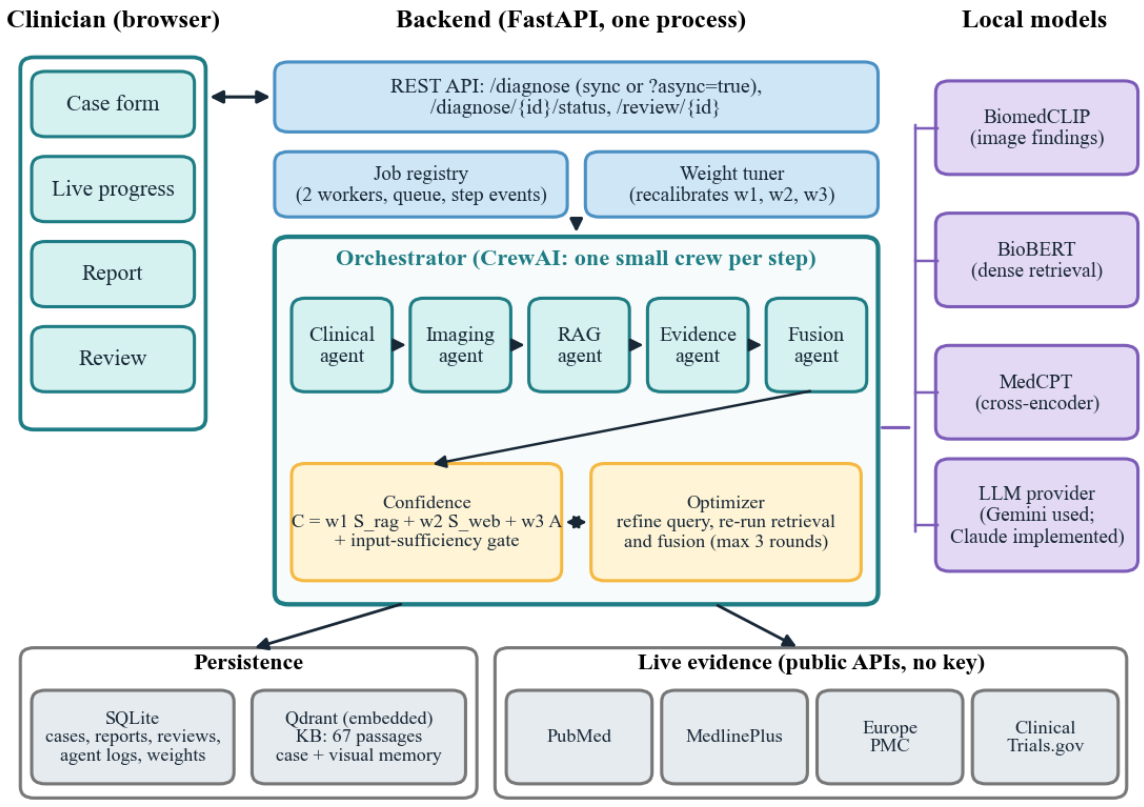



Figure 5.1: System architecture

Five principles shaped the design. **(1)** The language model reasons, but deterministic code scores: the model's self-reported certainty is ignored. **(2)** Every diagnosis must cite evidence that was actually provided; invented citations are rejected. **(3)** Every step degrades instead

of failing the run, and the degradation is visible. **(4)** A clinician stays in the loop, and review is mandatory below the threshold. **(5)** The system states its limits, such as unvalidated thresholds and unverified labels, wherever they apply.

## 5.2 Proposed approach and procedure

### 5.2.1 Pipeline overview

Figure 5.2 gives the workflow. Clinical extraction and imaging run first. Their output forms the retrieval query; offline retrieval and live evidence then run, and fusion produces the differential. A deterministic confidence follows. If it is below 0.65, or the case has no clinical anchor, the system either refines (at most three rounds) or hands the case to a clinician; a clinician's decision is stored and may later recalibrate the weights.

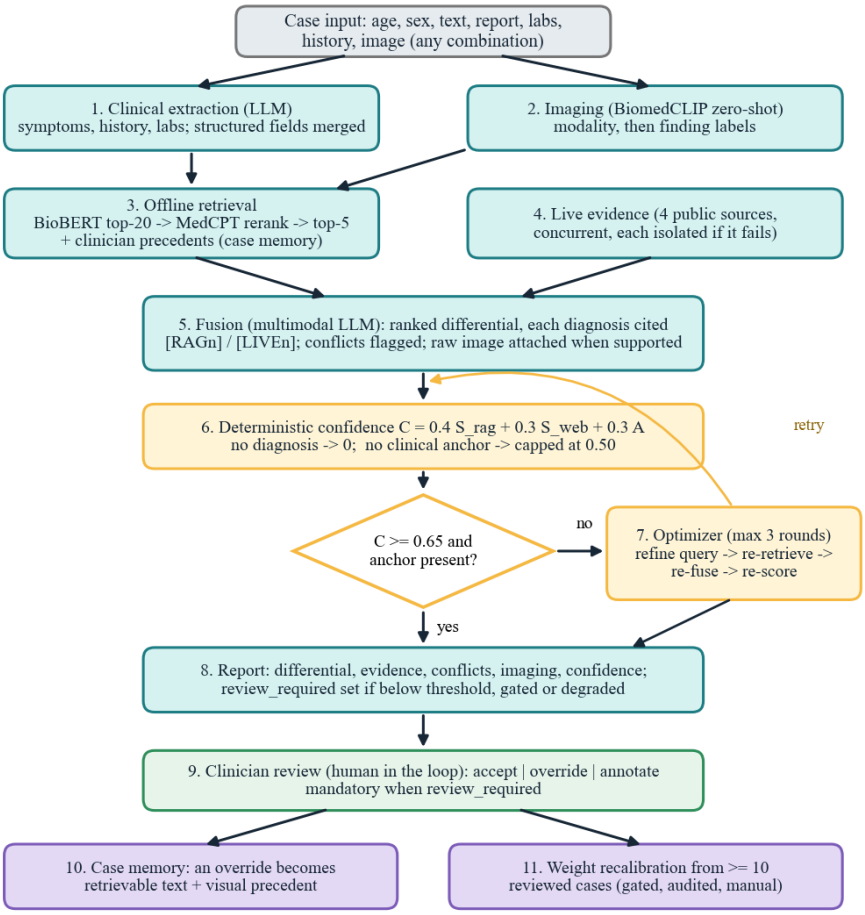



Figure 5.2: Pipeline workflow with the confidence-driven refinement loop and clinician review

### 5.2.2 Clinical extraction and structured fields

A language-model call extracts symptoms, history, laboratory values and imaging findings stated in the text into a schema-validated record, with one retry on malformed output. The case's typed history and labs are then merged in: labs override same-named extracted values,

and history is appended and de-duplicated ignoring case. This merge fixed a defect found during the project, in which the API stored these fields but the pipeline never used them (17 of the 31 test cases carry them). The prompt asks the model to keep the recent event that explains the complaint and its timing in the history (for example “collision during sport”) and to record stated negatives as “absence of X”; a stated negative alone does not count as a clinical anchor.

### 5.2.3 Knowledge base, retrieval and reranking

The offline knowledge base holds 67 verbatim passages (47 from MedlinePlus and 20 from PubMed; Table D.1), built by a script that fetches each source, keeps the Symptoms section first, and limits a passage to 340 words so that one passage remains one chunk. The query joins symptoms, “history of …” and imaging findings. Stage 1 retrieves the twenty nearest passages (ten while the knowledge base held 42) by BioBERT sentence-embedding cosine similarity (768 dimensions) from Qdrant. Stage 2 scores each query-passage pair with the MedCPT cross-encoder [8] and keeps the top five (the BioBERT backbone is described in [13]). If all five scores fall below 0.3, the query is broadened once to the first two symptoms.

Table 5.1: Composition of the knowledge base

| **Source** | **Passages** | **Content** |
|---|---|---|
| MedlinePlus Medical Encyclopedia | 22 | Condition articles: headache, hypertension, dementia, pneumonia, pulmonary edema, atelectasis, ... |
| MedlinePlus health topics | 25 | Topic overviews, e.g. hydrocephalus, heart failure, brain tumors, meningitis, chest pain |
| PubMed abstracts | 20 | Guidelines, consensus statements and reviews, e.g. BPPV (2017), normal pressure hydrocephalus (2021), chest pain (2021) |
| Total | 67 | 11 guideline-type and 56 symptom–disease passages; all verbatim and citable |

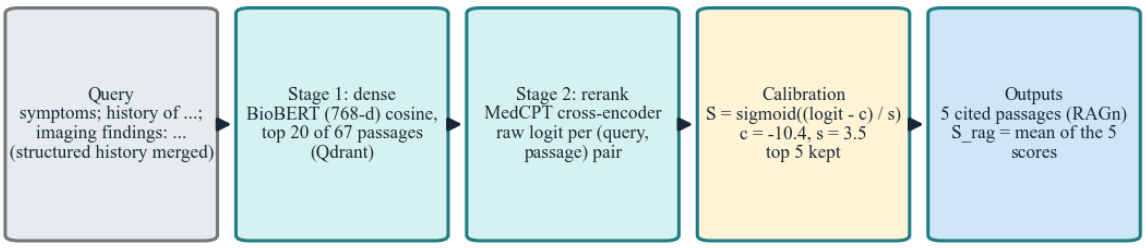



Figure 5.3: Two-stage retrieval and calibrated scoring

### 5.2.4 Score calibration

The confidence formula uses the reranker score as $S_{rag}$, so the raw logit must be mapped to 0–1. The mapping is a monotonic rescale, S = sigmoid((logit − c)/s), which never changes the ranking. For MedCPT the scale is s = 3.5 and the centre c was fitted by hand so that the mean $S_{rag}$ over 14 real queries (0.659) matches the earlier model's: c = −12.5 on the original 42-passage knowledge base, refitted to c = −10.4 when it grew to 67 passages and the candidate pool to 20 (off-topic queries: 0.284 before, 0.209 after). This is a prototype calibration, in the spirit of temperature scaling [10] but fitted to match averages, and it is not a validated probability. An optional ordering of the final five by reciprocal rank fusion [11] of the dense and cross-encoder ranks exists (`RERANK_ORDER=rrf`); it never changes which five passages are kept, so $S_{rag}$ is unaffected.

### 5.2.5 Live evidence

PubMed, MedlinePlus, Europe PMC and ClinicalTrials.gov are queried concurrently with a 10 s per-request timeout, one retry and a 30 s per-source deadline; a failure is isolated and recorded, near-duplicate titles are dropped, and the report states "live evidence unavailable" only if all four fail. Each item receives an evidence level (high, moderate, low) from a keyword heuristic on its text and publication type, which feeds $S_{web}$. The heuristic is crude and unvalidated.

### 5.2.6 Imaging

BiomedCLIP [6] scores the image against four modality labels, then against a finding label set for the detected modality (13 labels for brain MRI, 10 for CT, 12 for chest X-ray) using a softmax over cosine similarities scaled by 100. The top label is always reported; another is reported only if it exceeds 0.15 and is at least 60% as strong as the top, which removed a

spurious “rib fracture” that was seen on a real pneumothorax film. An image of any other modality is reported as “not assessed”, and no label from another modality is substituted. The finding labels are embedded as short descriptive captions (for example “head CT scan showing intracranial hemorrhage, hyperdense blood”) chosen by a held-out experiment (Section 6.3.4); the returned finding is still the plain label. A read whose top label scores below 0.40 is reported as *inconclusive*: the labels amount to “the classifier could not tell”, so they are shown for transparency but kept out of the retrieval query, the optimizer and the fusion prompt (which is told not to use them). The threshold is a chosen value, justified in Section 6.3.9. A confident read can also be checked against the multimodal model’s own reading of the raw image (off by default, Section 6.3.11); if the two disagree, the read is marked *contested* and treated like an inconclusive one.

### 5.2.7 Fusion with citations

The fusion call receives the clinical findings, the imaging labels, the retrieved passages tagged [RAG*n*], live items tagged [LIVE*n*] and precedents tagged [MEM*n*] or [VIS*n*]. It must return a ranked differential in which each diagnosis cites only IDs it was given, plus any conflicts between sources. The prompt treats imaging labels as hypotheses, forbids ranking a diagnosis on an imaging label alone, and forbids inventing a mechanism such as trauma. When an image is present, its modality is supported and the provider accepts images, the raw image is attached to the call; otherwise the path is text-only, and the choice is logged.

### 5.2.8 Confidence, the sufficiency gate and refinement

Confidence is computed outside the model as shown in Table 5.2:

$$C = w_1 \cdot S_{rag} + w_2 \cdot S_{web} + w_3 \cdot A \qquad (5.1)$$

Table 5.2: Parameters of the confidence score

| Symbol | Meaning | Value and source |
|---|---|---|
| $S_{rag}$ | Mean calibrated reranker score of the top five passages | sigmoid((logit + 10.4)/3.5); hand-fitted |
| $S_{web}$ | Mean evidence-level weight of live items | high 1.0, moderate 0.6, low 0.3; 0 if none |
| A | Agreement: 1 − 0.25 per flagged conflict, floored at 0 | Fixed penalty; unvalidated |
| $w_1$, $w_2$, $w_3$ | Weights | 0.4, 0.3, 0.3 prototype defaults; recalibration possible |
| Threshold | Below it the case is refined, then reviewed | 0.65, fixed, never recalibrated |
| Gate | No diagnosis; no clinical anchor; or only general symptoms | Confidence 0; capped at 0.50; capped at 0.60 (the last two with review required) |

The gate exists because the formula measures how much evidence was retrieved, not whether the case gave the system anything to reason from, and retrieval always returns something. A *clinical anchor* is at least one symptom, one laboratory value or one finding extracted from a report; history, demographics and the imaging model's own findings do not count. Without an anchor, more retrieval cannot help, so the optimizer is skipped. A second, narrower rule applies when every symptom is a general one (tiredness, feeling unwell, lightheadedness) and there are no laboratory values or report findings: confidence is capped at 0.60, review is required and the optimizer is skipped. It matches whole symptom phrases against a small hand-written list, so "arm weakness" is not caught; it is a transparent heuristic, not a validated classifier, and the 0.60 cap is a chosen value. Otherwise, below the threshold, the optimizer picks a strategy (expand, narrow or paraphrase), refines the query, re-runs retrieval, live evidence, fusion and scoring, keeps the best result, and stops at the threshold or after three rounds, after which the case is flagged for review. The report returns the evidence of the round that produced the best result, because that round's fusion cites its own numbered lists.

### 5.2.9 Clinician review, case memory and weight recalibration

The review screen offers accept, override and annotate with an optional one-to-five rating. An override is embedded and stored as a text precedent (retrieved on later cases only if BioBERT cosine similarity is at least 0.6) and, for supported images, a visual precedent (BiomedCLIP similarity at least 0.85, same modality only); both reach fusion framed as precedents to weigh, not as answers. Weight recalibration refits the three weights, with a floor of 0.1 and a sum of one, from accepted and overridden cases, only once ten distinct cases have been reviewed, writes an audit row, and is never run automatically.

### 5.2.10 Asynchronous API and interface

A case takes from about 30 seconds to several minutes, so `POST /diagnose?async=true` queues the run in a worker thread and returns at once, and the client polls `GET /diagnose/{id}/status` (Figure 5.4). The orchestrator reports each step, with its result and measured duration, and the optimizer reports each round; a failed step never carries the raw provider error. Every neural-network call (BioBERT, the reranker, BiomedCLIP) runs on one persistent thread, a workaround for a native crash described in Section 6.3.12. State is in the process's memory, concurrency is capped at two, and the synchronous endpoint is unchanged. The interface (Figure 5.5) has four screens: case entry with history and lab fields, live progress, the report, and review. The report shows a confidence gauge with the 0.65

threshold marked, citation chips that jump to the cited evidence, and a notice when the sufficiency gate applied.

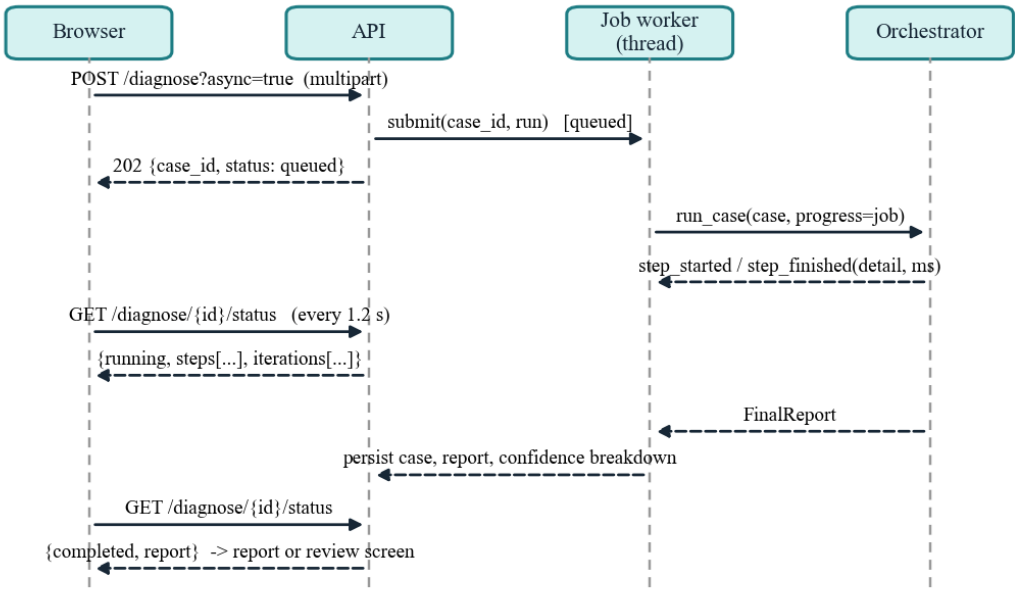
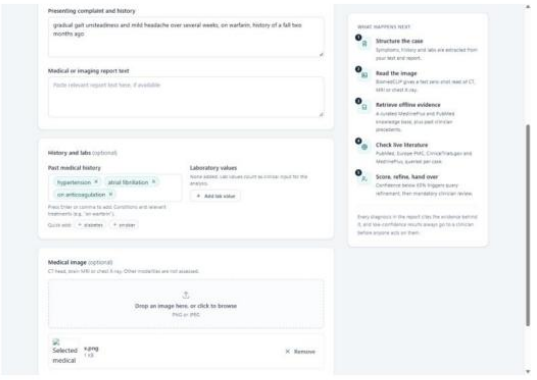
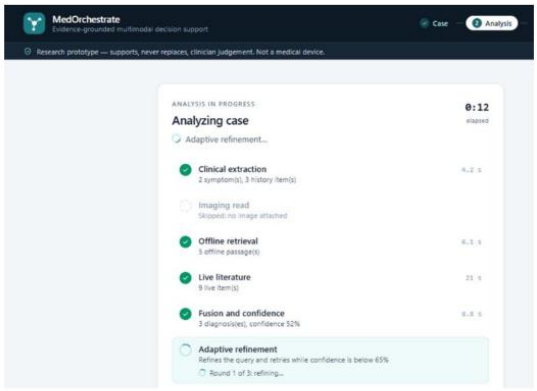



Figure 5.4: Asynchronous diagnosis: submit, progress events and polling

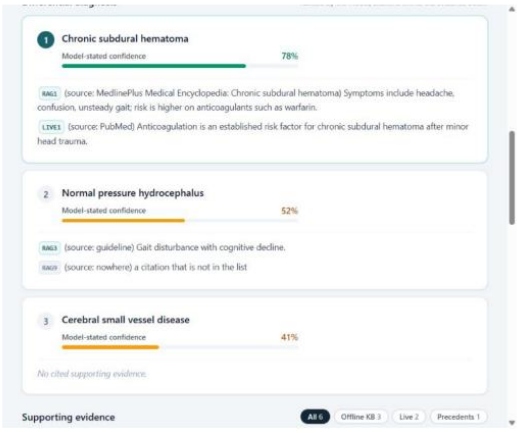
(a) Case form: history chips, lab rows



(b) Live progress (scripted mock run)




(c) Differential with citation chips (mock data)




(d) Review screen with the gate notice

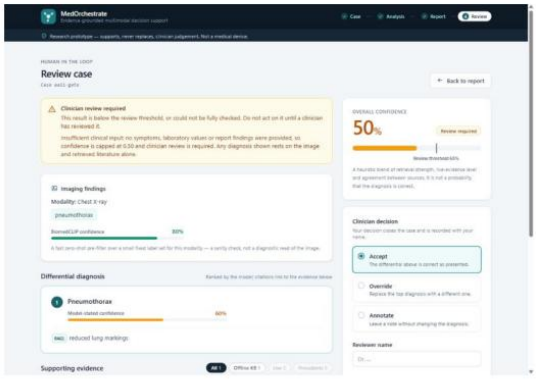



Figure 5.5: The interface: (a) case form with history and lab fields; (b)–(d) screens populated by a mock API, so their contents are illustrative and not system output

# Chapter 6
# Results and Analysis

## 6.1 Evaluation setup

Six instruments were used. **(1)** The automated test suite. **(2)** A retrieval harness that runs the real BioBERT and reranker pipeline on an in-memory index of the committed corpus, over 14 queries recovered verbatim from the live run's log, so they are the queries the agents actually built (not hand-written paraphrases). **(3)** A real-image benchmark of 62 films, later extended by two sets of fresh ones (17 and 9). **(4)** A held-out experiment on prompt wording. **(5)** A live run of 31 cases through the real pipeline with the Gemini model, in an isolated copy of the database and vector store so that the real case memory was untouched. **(6)** Browser verification of the interface.

Safeguards against invalid comparisons: the prompt experiment's split, candidates and decision rule were committed before any result was seen (commit `9049e41`), and its test half was evaluated once, with the script refusing a second run; every proportion carries a Wilson interval and a chance baseline; analyses exclude images flagged for possible training overlap or carrying annotations; and results that came from a degraded run (a provider-quota error) are marked invalid, not averaged in.

## 6.2 Evaluation criteria

**Retrieval.** hit@*k* is the share of queries with a relevant passage in the top *k*; MRR@5 is the mean of 1/rank of the first relevant passage (0 if none in the top five). Relevant passages are the developer's judgement, not a clinician's. **Imaging.** Top-1 and top-3 accuracy of the finding label given the true modality, modality-detection accuracy, the share of images for which the pipeline's reported findings contain the truth or include a wrong finding, and sensitivity and specificity of normal versus abnormal. Proportions use the Wilson score interval [15], $(p + z^2/2n \pm z\sqrt{(p(1-p)/n + z^2/4n^2)}) / (1 + z^2/n)$ with $z = 1.96$; chance is the expected accuracy of a random ranking, and the majority baseline always predicts the commonest label. Paired comparisons use an exact McNemar test. **System.** Review rate, confidence, optimizer rounds, and whether input that should lead to abstention (an empty or image-only case) was sent for review.

## 6.3 Result

### 6.3.1 Automated tests

The backend has 641 automated tests and the frontend 124 (23 helper tests, 94 component tests and 7 browser end-to-end tests). Counting non-blank lines, the backend test code is about 9,100 lines against about 5,900 lines of backend code; the frontend has about 2,500 lines of code and 1,200 of tests. They cover the agents, the confidence and gate logic, the orchestrator, the API and the job registry, storage, the weight tuner, retrieval and reranking, imaging and the evaluation tooling, with external services faked; the component tests (React Testing Library) cover the case form, report, review and progress screens with the API mocked, and not layout; the 7 end-to-end tests drive Chrome against the built frontend and a real backend whose models and network calls are faked, covering validation, an ordinary case with a citation jump, the vague and image-only notes, the optimizer's rounds on screen, an override and a reload, and test the browser-to-backend boundary, not any model. Scoring and safety logic was checked by deliberate mutation: removing the confidence gate, the failure detection in progress reporting, the hiding of provider errors, the reranker ordering and the top-3 scorer each made a specific test fail. These tests show the code does what it was written to do. They are not evidence that the clinical behaviour is right.

### 6.3.2 Retrieval

Table 6.1: Retrieval on 14 real agent queries (hit@5, hit@1 and MRR@5); 42-passage knowledge base unless stated

| **Pipeline** | **hit@5** | **hit@1** | **MRR@5** |
|---|---|---|---|
| Before: history and labs absent from queries; MS MARCO reranker | 13/14 | 7 | 0.685 |
| + structured history and labs merged into the findings | 14/14 | 8 | 0.768 |
| + reciprocal rank fusion ordering of the top five (MS MARCO) | 14/14 | 10 | 0.827 |
| MedCPT reranker, cross-encoder order, 42-passage knowledge base | 14/14 | 11 | 0.863 |
| MedCPT with reciprocal rank fusion ordering (not default) | 14/14 | 12 | 0.895 |
| Reference: dense retrieval only, no reranker (42 passages) | 13/14 | 9 | 0.774 |
| **Knowledge base grown to 67 passages; candidate pool 20; score centre refit (current)** | **14/14** | **9** | **0.764** |

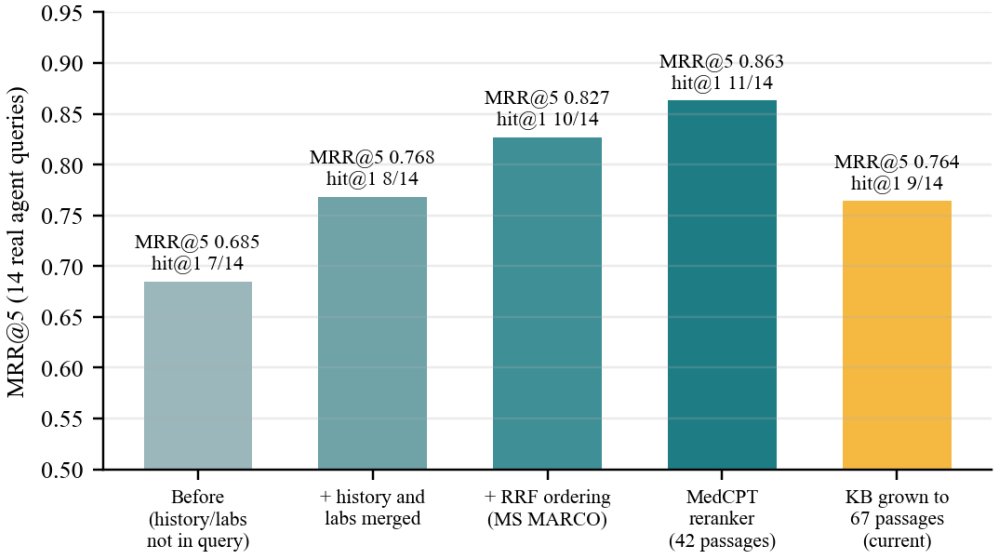



Figure 6.1: Retrieval quality as the pipeline was corrected, and after the knowledge base grew

Two corrections account for most of the gain. First, the API stored the typed history and labs but the pipeline never used them; merging them removed the one miss (a case whose query lacked its anaemia history) and lifted MRR@5 from 0.685 to 0.768. Second, MedCPT [8] replaced the general-domain MS MARCO cross-encoder: it stopped off-topic passages from topping generic queries (pulmonary tuberculosis had ranked first for a child's headache) and lifted MRR@5 to 0.863. Reciprocal rank fusion [11] of the dense and cross-encoder ranks added a little with MedCPT (0.895) but moved the subdural-hematoma passage for the anticoagulated-fall case to the edge of the top five, so it is off by default. Earlier, a fault in the knowledge-base builder (it cut off the Symptoms section in 10 of 15 encyclopedia articles) had hidden that passage; fixing it took a hand-written 14-query check from 12/14 to 14/14.

**Cost.** MedCPT is not better everywhere: the cough-and-smoker case fell from first to fourth, and the subdural passage is third, not second. Replacing the reranker changes confidence, so the scale was refitted so that the mean $S_{rag}$ over the 14 queries stayed 0.659; individual queries still moved by up to 0.29 (0.12 of confidence). Fourteen queries with developer-judged relevance are a regression check, not a benchmark.

**Knowledge-base growth.** The answer key and the 31 live reports named diagnoses with no passage behind them, and the imaging models can name findings (pulmonary edema, atelectasis, mediastinal widening, mass lesion, vascular malformation, midline shift, calcification) that the knowledge base did not cover. Twenty-five passages were added (MedlinePlus topics and articles, six PubMed abstracts and the 2021 AHA/ACC chest-pain guideline), taking it from 42 to 67. This helped coverage and hurt ordering. For 21 queries

written for the new conditions (by the person who added the passages, so a coverage check and not an independent test), a relevant passage reached the top five for 18 against 3 on the original 42 passages; the misses were mediastinal widening, vertebrobasilar insufficiency and musculoskeletal chest pain. On the 14 original queries hit@5 stayed 14/14 but MRR@5 fell from 0.863 to 0.764 and hit@1 from 11 to 9, because new and plausibly related passages (long-COVID headache for a child with a cold, a posterior reversible encephalopathy review for a memory-and-gait case) now rank above the one the developer had marked relevant; those relevance sets were not changed. At the old pool of ten candidates the anticoagulated-fall case lost its subdural-hematoma passage altogether, so the pool is now twenty, and the score centre was refitted (−12.5 to −10.4) so that the mean $S_{rag}$ of the 14 queries is 0.659 again. A re-judging of the 14 relevance sets (every plausibly relevant passage among each query's 20 candidates added, written down before re-scoring, by one non-clinician) shows that about 40% of the drop was a stale key: MRR@5 is 0.881 on 42 passages and 0.821 on 67 under the re-judged sets, against 0.863 and 0.764 under the original ones; the rest is real, related passages ranking above the marked one. Confidence values from before this change are therefore comparable with later ones only on that average.

### 6.3.3 Real-image imaging benchmark

The earlier imaging benchmark scored BiomedCLIP on generated gradient pictures with an arbitrary intended label, so near-chance was guaranteed and it measured nothing; it was deleted. The replacement has 62 real images (32 chest X-ray, 18 head CT, 12 brain MRI) in 60 groups: 56 downloaded from Wikimedia Commons under open licences and 6 already in the repository. Each carries a finding label taken from its Commons caption, not from a radiologist. A contact-sheet review found and replaced about two dozen images with arrows, circles or boxes drawn on the film, composite panels, viewer screenshots, or visible patient-identification text.

Table 6.2: Zero-shot finding accuracy on real images (true modality given; Wilson 95% interval)

| **Modality** | **Images** | **Top-1, bare labels** | **Top-1, descriptive captions** | **Top-3, captions** |
|---|---|---|---|---|
| Chest X-ray | 32 | 17/32 (53%; 36%–69%) | 20/32 (62%; 45%–77%) | 25/32 (78%; 61%–89%) |
| Head CT | 18 | 6/18 (33%; 16%–56%) | 8/18 (44%; 25%–66%) | 14/18 (78%; 55%–91%) |
| Brain MRI | 12 | 3/12 (25%; 9%–53%) | 7/12 (58%; 32%–81%) | 10/12 (83%; 55%–95%) |
| **All images** | **62** | **26/62 (42%; 30%–54%)** | **35/62 (56%; 44%–68%)** | **49/62 (79%; 67%–87%)** |

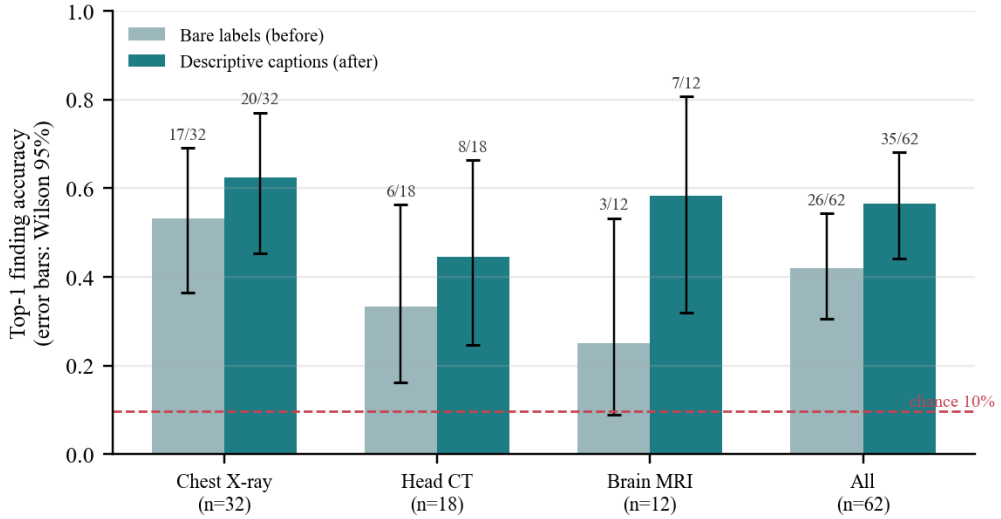



Figure 6.2: Top-1 finding accuracy before and after the caption change, with Wilson intervals

Against chance (10%) and an always-the-commonest-label baseline (21%), the zero-shot model is clearly better than random but not good: with the original labels about one chest film in two and one CT in three got the right top label. Modality detection was 58/62 (94%); four of 18 CTs were detected as MRI. The pipeline's reported findings contained the truth for 28/62 (45%) of images and included at least one wrong finding for 39/62 (63%) (original labels). The most consequential failure was head-CT hemorrhage: four of five hemorrhage CTs were called acute infarct (three at scores of 0.95 or higher) and the fifth normal. Five abnormal images were called normal. The imaging read is therefore a coarse pre-filter, not a diagnostic result.

### 6.3.4 Held-out test of prompt wording

Because 62 images are too few to try several wordings and keep the best, the design was fixed first: a group-level split (37 development and 25 test images, seed 20261113), five finding-prompt and four modality-prompt candidates, selection by development MRR, and adoption only if the test half gained at least two top-1 images without lowering MRR. On the development half the caption "… showing …" scored 23/37 against 19/37 for the bare label. On the test half, used once, it scored 12/25 against 7/25 (MRR 0.61 against 0.49; seven images gained and two lost; exact McNemar p = 0.18, not significant). The rule therefore adopted it; modality wording gained one image, below the bar, and was not adopted.

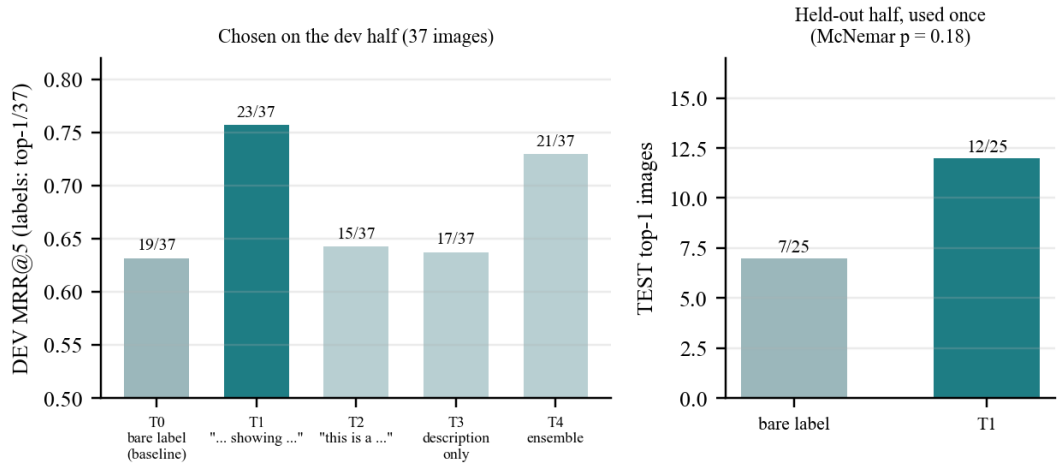



Figure 6.3: Prompt-wording experiment: development choice and the one-time held-out result

The change has a cost the adoption rule did not see. Specificity (normal called normal) fell from 7/9 to 5/9 and sensitivity rose from 48/53 to 51/53; the effect is concentrated in one failure mode, in which a “sinus abnormality” caption attracts normal brain images, and it appears on the development half and not on the held-out half (1/3 to 2/3, three normals). Hemorrhage CT improved from 0/5 to 3/5, no longer called infarct. With nine normal images from about six subjects, a real specificity cost is not excluded. The change is kept, with one environment variable to reverse it, and the trade-off is recorded in the project documentation. The next subsection tests it again on images that played no part in the choice.

### 6.3.5 Extension sets: further tests on fresh images

After the prompt choice was made and its held-out half spent, 17 more real images were added to fill gaps: four normal chest films, two CT films with midline shift, one more CT infarct, two more slices of the one normal head-CT subject, and eight brain MRIs (three slices of one vascular malformation, a chronic infarct, edema, white-matter hyperintensity and two normals). Their plan was written and committed before any of them was scored, they were kept out of the split and the frozen experiment, and nothing was re-tuned afterwards. Atrophy and CT calcification stay uncovered, because every usable candidate was a composite panel or had arrows drawn on it.

Table 6.3: The 17 fresh images under the current captions and the bare labels

| **Measure** | **Current captions** | **Bare labels** |
|---|---|---|
| Top-1 finding accuracy | 10/17 (59%; 36%–78%) | 11/17 (65%; 41%–83%) |
| Top-3 | 15/17 | 12/17 |
| Normal films called normal (specificity) | 6/8 | 8/8 |
| Abnormal films flagged (sensitivity) | 8/9 | 6/9 |
| Modality detected | 16/17 | 16/17 |

Chance is 11% and always predicting the commonest label gets 53%, so a top-1 near 60% is not impressive here. Paired image by image, the captions gained 3 and lost 4 (exact McNemar p = 1.0): **on fresh images the captions were no better than the bare labels**, and the advantage seen on the original held-out half was not reproduced. Neither result is significant, and 17 images cannot show a difference of this size in either direction. Two of the new images share a subject with the original normal head CTs, and the edema image is flagged for possible training overlap, so even this set is not fully independent.

A second set of nine images (plan again committed before scoring) targeted what was still missing: four CT calcification films, one MRI with cerebral atrophy, a second normal head CT, two normal brain MRIs and a lateral chest film. The captions got 5/9 top-1 against 3/9 for the bare labels (3 gained, 1 lost, p = 0.63). Calcification went three of four with the captions and none with the bare labels, the opposite of the expectation recorded beforehand; four films, two of them the same age-related pattern, are a signal and not a result. The normals went badly again: the normal head-CT slice through the orbits was called “sinus abnormality” at 0.98, a normal sagittal MRI “cerebral atrophy”, and the lateral chest film (a view the label set was not written for) “pneumothorax”. Atrophy is still covered by a single film and CT atrophy not at all, because every other candidate was a composite panel, carried arrows or showed the wrong kind of atrophy.

Over both fresh sets (26 images) the captions scored 15/26 against 14/26 (6 gained, 5 lost, exact McNemar p = 1.0). What holds is the trade-off: they flag more abnormal films (sensitivity 12/14 against 9/14) and call more normal films abnormal (specificity 7/12 against 10/12). Over all 88 images the top-1 totals are 50/88 against 40/88, but that figure includes the tuning set and is not an unbiased estimate. The captions therefore stay, with a weaker claim: more sensitive, less specific, and a net accuracy gain that fresh images did not confirm. Which wording is better for this project depends on whether a missed abnormality or a false alarm is the costlier error, which is a clinical judgement that 88 images cannot settle.

### 6.3.6 Live run of 31 cases

All 31 cases (24 synthetic and 7 real-image) returned a report through the real pipeline with `gemini-3.5-flash-lite`. This run used the version *before* the input-sufficiency gate, the history-and-labs fix, the MedCPT reranker and the new image captions, so it describes an earlier system. Mean confidence was 0.71 (excluding CASE007, which failed on a provider-

quota error and is invalid); 3 of 31 cases were sent for review; optimizer rounds were 0 for 17 cases, 1 for 11, 2 for 1 and 3 for 2; live evidence was present for 28.

The run exposed two flaws, both visible in Figure 6.4. An empty case (CASE013) came back with no diagnosis and confidence 0.79, above the threshold, so it skipped review: the formula scored the evidence the optimizer had retrieved and not the absence of a diagnosis. And of the six cases whose correct behaviour is to abstain (empty or image-only input), only one was sent for review; the others reported confidence of 0.67 to 0.79. The agreement signal was 1.00 for almost every case, so about 0.30 of the score was free, and retrieval always returns something. Both were fixed (no diagnosis gives confidence 0; no clinical anchor caps it at 0.50 and skips the optimizer), with unit tests. The fix could not be re-run live at first because the provider quota was exhausted again (four attempted cases failed safely: confidence 0, review required); it was confirmed later, below. A vague but non-empty case (CASE008) received 0.68; a later rule (Section 5.2.8) now caps a case whose only symptoms are general at 0.60 and sends it for review, as the re-run below confirms.

**The same 31 cases on the final system.** After the provider answered again, all 31 cases were run through the real pipeline with the final code, the 67-passage knowledge base and the new image captions, on an isolated copy of the store. Every case returned a report and there were no quota errors; the backend process crashed once (a native exit with no traceback) and the nine unfinished cases were re-run. Mean confidence was 0.63 (all cases valid this time); 13 of 31 cases were sent for review, against 3 before; **all 6 of the six abstention cases went to review**, against one; the empty case returned no diagnosis and confidence 0.00 where it had scored 0.79; the image-only cases were held at 0.50 and the vague case (CASE008) at 0.60, each with its note; optimizer rounds were 0 for 21 cases, 1 for 2, 2 for 1 and 3 for 7; live evidence was present for 30. The real pneumothorax film with no text (REAL007) now goes to review at 0.50, the intended cost of the gate. In none of the reports does a diagnosis cite an evidence item that is missing from its evidence list, including the ten that entered the optimizer (one of 14 did in the first run); this shows that nothing dangles, not that the winning round's evidence was the one returned. Thirteen of 31 reviewed is a property of the current threshold and weights, which are unvalidated.

### 6.3.7 LLM-judged consistency check

The 31 final reports were judged by a language model on four measures defined as in RAGAS [14] but implemented natively: whether any top-three diagnosis matches the answer key (abstention cases are scored by rule: correct if flagged for review), faithfulness (do the

passages a diagnosis cites support it: supported 1, partial 0.5, unsupported 0), context precision (average precision over the retrieved passages judged relevant) and unsupported claims. The judge was `gemini-3.5-flash-lite`, the same model family that wrote the reports, so it tends to favour them; it scored every metric (no judge errors) except the two that cannot apply to the empty case.

Table 6.4: Judge results over the 31 final reports (a consistency check, not accuracy)

| **Measure** | **Result** | **Scored** |
|---|---|---|
| Diagnosis correct against the answer key | 0.903 (28 of 31) | 31 (6 by rule) |
| Faithfulness of cited evidence | 0.453 | 30 |
| Context precision | 0.616 | 30 |
| Cases with unsupported claims | 1 (3 claims) | 31 |

The three diagnoses marked wrong were inspected. CASE004 is a real failure: the picture is a generated pattern, the zero-shot read returned “edema, cerebral atrophy, chronic infarct”, and the differential followed it (transient ischemic attack, vertebrobasilar insufficiency, posterior reversible encephalopathy) at confidence 0.73 with no review, although the key says the text decides; an image of no clinical value can still steer the answer. REAL005 is defensible: the film is normal and the system, correctly, named no lung disease, but it also did not name the musculoskeletal cause the key expects. CASE007 looks like a judge error: the differential's second entry, “hypertension-related dizziness / chronic daily headache”, matches the key’s “hypertension-related headache”. Faithfulness of 0.45 is low, and most of the image-only cases scored 0.00; this was not checked against a clinician, so it is not known whether the judge is strict or the citations are weak. The numbers are small, from one judge, and they are not clinical validation.

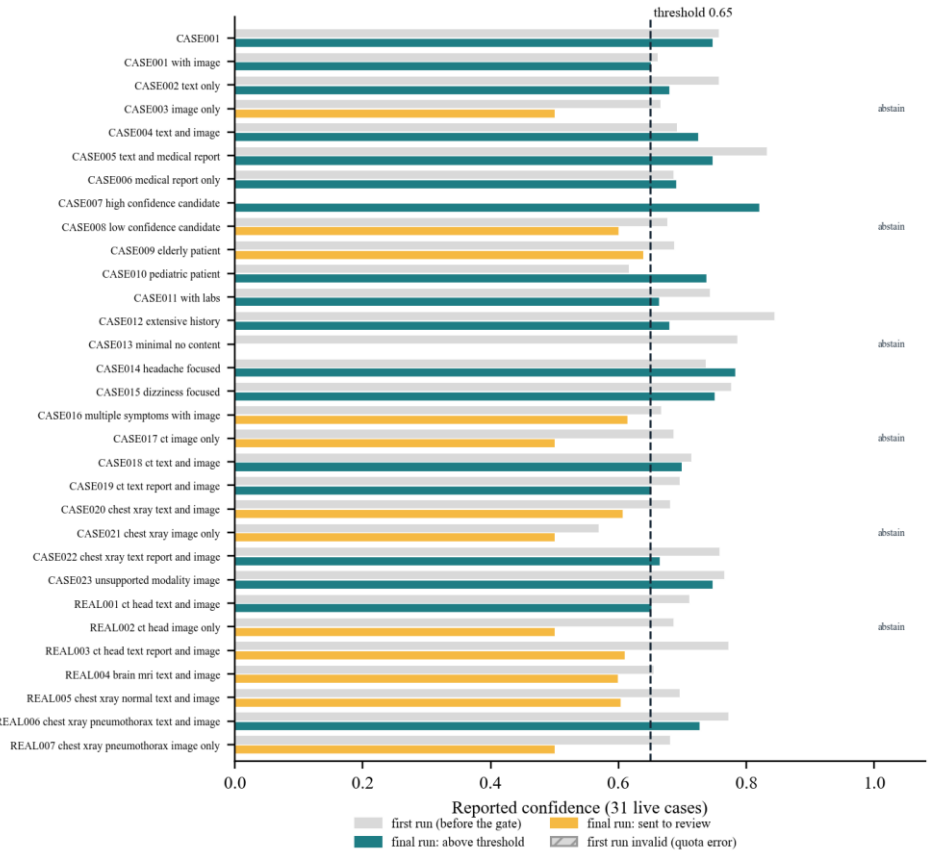



Figure 6.4: Reported confidence for the 31 live cases: first run (before the input-sufficiency gate, grey) and the run on the final system (coloured)

### 6.3.8 A second pass over the 31 cases: run-to-run variation

The 31 cases were run a second time on the final code (with the inconclusive-image rule, the extractor prompt fix and the single model thread), against the same store and settings. This run finished without a crash. Mean confidence moved by 0.023 per case (median 0.01, largest 0.17); 27 of 31 stayed within 0.05 and 30 within 0.10. The review decision was the same for 27 of 31 cases; all four that changed sat next to the threshold (0.58 to 0.69) and crossed it, so the number sent to review went from 13 to 15. The top diagnosis was the same (by a loose name match) in 22 of 31 cases, on average 83% of the first run's top three reappeared in the second, and the number of optimizer rounds was the same in 26 of 31. The six abstention cases were identical in both runs (the rules are deterministic): 0.50, 0.60 and 0.00 with review. So the model-driven parts vary by a few points of confidence and about a quarter of top diagnoses, and a case that sits near 0.65 can fall on either side of the review line from one run to the next; the clearly high and the clearly gated cases do not. CASE004 was the only case whose image read was marked inconclusive, as intended. Two runs are not a distribution: this gives the size of the variation, not its interval.

**Rehearsal through the interface.** With the real backend and Gemini, two cases were run from the browser on the final code. The hypertension-and-headache example streamed its real steps with timings and returned a report at 76% above the threshold, with cited evidence; the elderly-on-warfarin example showed the optimizer's rounds live (round 1 at 53%, then round 2 refining) and ended on the review screen at 62%, chronic subdural hematoma first, with the notice that clinician review is required. These are two cases watched by the developer, not a usability study; one brain-MRI case in an earlier run took 505 s with three optimizer rounds, too slow for a live demonstration.

### 6.3.9 A weak image read steering the differential, and the fix

The judge run exposed one genuine failure (CASE004): a generated picture was paired with "headache and dizziness, history of hypertension", BiomedCLIP scored its three labels 0.34, 0.23 and 0.22, and listing them as findings pulled retrieval and the differential toward transient ischemic attack, vertebrobasilar insufficiency and posterior reversible encephalopathy at confidence 0.73 with no review. A flat score distribution means the classifier could not tell, so reads whose top label scores below 0.40 are now marked inconclusive and kept out of the query, the optimizer and the fusion prompt. Re-run live, CASE004 returned vestibular migraine, hypertension-related headache and dizziness, and giant cell arteritis at 0.75, driven by the text; four control cases with decisive reads (0.65 to 0.82) were unchanged in how their images were used. The rule has limits. It cannot catch a confident wrong read: a generated brain MRI scores "cerebral atrophy" 0.65 and a generated chest film "pulmonary nodule" 0.9, because a softmax over a closed list looks decisive on nonsense; a second reading of the image, described below, addresses part of this. The threshold is not validated: on the 88 real images a top score under 0.4 was right 4 times in 9 and a score of 0.8 or more 24 times in 35, so the rule discards about four correct reads to avoid five wrong ones. It is a safety measure for indecisive reads, not an accuracy gain.

### 6.3.10 The multimodal model's own reading of the image

The fusion step also sees the raw image, so its own reading is a second, previously unmeasured imaging signal. The same closed-set question BiomedCLIP is asked (the true modality given; choose the best label from the same list) was put to the multimodal model for each of the 88 benchmark images, one call per image.

Table 6.5: Raw-image reading by the multimodal model against BiomedCLIP, same 88 images

| Measure | Multimodal model | BiomedCLIP (captions) |
|---|---|---|
| Top-1 finding accuracy | 68/88 (77%; 67%–85%) | 50/88 (57%; 46%–67%) |
| Right where the other was wrong | 26 images | 8 images |
| Abnormal films flagged (sensitivity) | 66/67 | 63/67 |
| Normal films called normal (specificity) | 20/21 | 12/21 |
| MRI / CT / chest X-ray | 17/23 · 23/28 · 28/37 | 14 · 14 · 22 |

Paired, the multimodal model was right on 26 images where BiomedCLIP was wrong and wrong on 8 where it was right (exact McNemar $p = 0.003$). This is probably optimistic: the labels are the uploaders', the images are public teaching examples that a large model may have seen in training, the closed list flatters it, and it is one model, one prompt and one pass. What it supports is sending the raw image to fusion, and caution about letting the zero-shot labels alone drive the query; it is not a validated imaging result.

### 6.3.11 A second reading of the image: the cross-check

The inconclusive-read rule cannot catch a confident wrong read, so a second check was added. When `IMAGING_CROSS_CHECK` is on, the multimodal model reads the raw image against the same closed label list; if its label is not among the labels BiomedCLIP reported (two "normal" labels count as the same), the zero-shot read is marked *contested*. A contested read is treated like an inconclusive one: it is shown to the clinician but kept out of the retrieval query, case memory, the optimizer and the fusion prompt, which is told only that two readings disagreed. It fails open (a provider without image input, an error or an unparseable answer leaves the read as it was) and costs one extra model call per image case. It is off by default, because every result above was produced without it, and on for the demonstration. Replayed on the 88 saved readings, with no new calls, 25 of the 29 wrong zero-shot reads with a top score of 0.40 or more disagreed with the multimodal model, and 5 of the 50 right ones did; this is optimistic for the reasons given above.

End to end, the 18 cases that have an image were run once with the cross-check on, beside a full run with it off (same code, store and demonstration captions), and both runs were judged. Table 6.6 lists the three reads it contested: two generated pictures and a real, normal T1 brain

MRI that BiomedCLIP calls “cerebral atrophy” at 0.79. It left the correct reads of the six real CT and chest films unchanged.

Table 6.6: Zero-shot reads contested by the second reading, with the cross-check off and on

| Case | Zero-shot read (score) | Second reading | Confidence | Review |
|---|---|---|---|---|
| CASE004 (generated picture) | acute infarct (0.49) | no significant abnormality | 0.72 to 0.67 | no, no |
| CASE022 (generated chest film) | pneumothorax (0.80) | pulmonary nodule or mass | 0.75 to 0.80 | no, no |
| REAL004 (real normal T1 MRI) | cerebral atrophy (0.79) | ventricular enlargement | 0.63 to 0.57 | yes, yes |

Over all 18 cases the review decision was the same, mean confidence changed by 0.013 (largest 0.066) and the top diagnosis was the same in 14. That is inside the 0.02 to 0.05 by which confidence moves between runs of the same settings, so these runs support a claim about the image label only, not about confidence or review: the text and the raw image that fusion also sees carry the result. The same-family judge scored the off run at 0.968 diagnosis correct (30 of 31), faithfulness 0.41 and context precision 0.611; on the 18 image cases the off and on runs differed by one case or a few hundredths, and the extra unsupported-claim cases in the on run concern clinical wording, none a contested case. The limits: one run per setting gives no spread; no clinician has confirmed that the contested labels were wrong; and the second reading can be wrong in the same way (both models read a generated chest film as “pulmonary nodule or mass”).

### 6.3.12 Stability: a native crash and its workaround

The backend process died, with no Python traceback, in three long runs (a native access violation, exit code 139). With `faulthandler` on, the trace pointed into a torch operation inside a worker thread, in the middle of the MedCPT cross-encoder. The orchestrator runs each step with `asyncio.run()`, which creates a new event loop and a new default executor, so retrieval ran on a freshly created thread almost every time. Repeating only the real retrieval, rerank and image-read code in that pattern, with no language-model call, crashed after about 35 iterations; with every neural-network call routed through one persistent thread, 300 iterations (33 minutes) completed. That points at per-thread native state in the numerical library rather than at one model, but the native cause was not found, so this is a measured workaround, not a root-cause fix; it also serialises model calls across concurrent cases.

### 6.3.13 The demo CT images

Two of the three real CT demonstration images (normal head slices at orbit and skull-base level, one subject) are read as "sinus abnormality" by the current captions (0.72 and 0.89), and one differential then lists sinusitis; with the bare labels all three read as normal at 0.99 or higher. This is the captions' specificity cost seen on the demonstration material. For the demonstration the bare labels are therefore used (an environment setting, as is the cross-check described below; the code defaults and the evaluation stay on the captions, with the cross-check off): re-run live with them, both CTs read as normal, sinusitis left the differential.

### 6.3.14 Review, case memory, recalibration and progress

With the real models and a temporary store, an override on an image case wrote one text and one visual precedent; a later similar case retrieved both, and an unrelated case retrieved neither. The round trip also exposed three defects that were fixed (the clinician's diagnosis was not being stored, there was no relevance cutoff, and visual precedents ignored modality). Weight recalibration did not run on real data: only 3 of the 10 required reviewed cases exist, so it correctly does nothing. A labelled simulation (seven developer-written reviews) showed the mechanism only; its output weights reflect a rule in the script, not clinicians, and must not be quoted as learned.

### 6.3.15 Interface verification

The four screens were verified in a browser against a recording mock API and, for failure paths, the real backend: the form sends the fields the backend expects, validation messages appear for faulty lab and history input, citation chips jump to their evidence, a new case stops the polling loop, and there is no horizontal overflow at 375 px. This was functional verification, not a usability study with clinicians.

**Container deployment.** The system was also built and run as containers on the development machine (Docker Compose: backend, frontend and a Qdrant server). The backend image installs the CPU-only builds of torch and torchvision (the image is 1.0 GB instead of 3.4 GB), and a volume keeps the roughly 2.4 GB of downloaded models, so a recreated container loads BioBERT, MedCPT and BiomedCLIP with the network blocked. After the knowledge base was ingested (67 passages, 3 minutes 50 seconds including the first model download), seven demonstration cases and one example case from the browser interface ran through it without error, and the three contested reads above reproduced. This was one machine, the

requirements are not pinned, and image upload from the browser was not driven there (the same cases went through the API).

## 6.4 Comparative analysis

Table 6.7: Results against baselines and expectations

| Measure | Baseline or expectation | Result |
|---|---|---|
| Retrieval, MRR@5 | 0.685 before the fixes; 0.774 dense only | 0.863 on 42 passages; 0.764 on 67 (hit@5 14/14 for both; hit@1 11 then 9) |
| Imaging top-1, original 62 | Chance 10%; commonest-label 21% | 42% with bare labels; 56% with captions (held-out half 12/25 against 7/25) |
| Imaging top-1, 26 fresh images | Chance about 11%; bare labels 14/26 | Captions 15/26 (58%); 6 gained, 5 lost, p = 1.0 |
| Imaging modality detection | Required for every image | 83/88 (94%) |
| Abstention on 6 empty, image-only or vague cases | Review required in all six | 1/6 before the gate; 6/6 in the re-run on the final system |
| Retrieval benefit in the literature | Up to 18% accuracy gain over chain-of-thought [4] | Not comparable: different task and metric |

The literature figures cannot be compared with these directly, since they measure answer accuracy on question-answering benchmarks while this project measures passage ranking and image findings on its own small sets. The honest comparison is each result against its own baseline, as in Table 6.7.

## 6.5 Discussion and Interpretation

The results support the engineering objectives (O1 to O3, O5, O6) and O4 in its mechanical sense (the gate and cap route the cases they were built for); they do not support O4's calibration. The pipeline runs end to end under every input combination and under provider failure; retrieval can be measured and improved; and imaging can be characterised honestly. The most useful findings were negative or corrective: a confidence formula built on retrieval volume was blind to missing input, a regression reported on a hand-written query was not present on the query the agent really built, and a prompt change that looked better overall hid a loss on normal images. Each was found by evaluating on real inputs with baselines and intervals, which is what the project learned to do.

What the results do not support is any claim of clinical accuracy. The confidence threshold and weights are unvalidated, the imaging labels are not radiologist-verified, and the answer key is not clinician-verified. The contribution is a system that makes its evidence and its

uncertainty visible and routes doubtful cases to a person, evaluated as far as the available quota and people allowed.

## 6.6 Limitations

- **Small samples.** 14 retrieval queries (relevance judged by the developer, for the 42-passage knowledge base), 88 images (21 normal, a single normal head-CT subject, one atrophy film), 25 held-out images, 26 fresh images, 31 live cases; intervals are wide and a McNemar test on 25 images can only detect very large effects.
- **Unverified labels.** The expected-diagnosis key, the retrieval relevance sets and the image labels are developer- or uploader-derived. None is clinician-verified.
- **Optimistic images.** Wikimedia Commons holds clear teaching examples; real films are harder. BiomedCLIP was pretrained on published articles, so overlap with some images cannot be excluded (a keyword heuristic flagged only three).
- **Three full runs, one model.** The final 31 cases were run twice on the default captions (the first had a backend crash and a re-run of the unfinished cases) and once more under the demonstration captions; the image cross-check was run once, on the 18 image cases. Three passes show the size of the run-to-run variation but are not a basis for confidence intervals.
- **Single provider, and a judge of the same family.** Every live run and the judge used one Gemini model; the Claude path was not exercised end to end, and a model judging its own output tends to favour it.
- **In-process jobs.** The asynchronous API assumes one server process; a restart drops running jobs; there is no cancel button.

### 6.6.1 Evaluation not yet performed

The following have **no results**: (1) judging by a model of a different family, or by a clinician; the one judge run above is a consistency check by the same family [12]; (2) evaluation by clinicians, of which only 3 reviews of the 10 required exist; (3) calibration of the 0.65 threshold, the weights and the 0.60 cap (13 of 31 cases going to review is a property of the current, unvalidated values); (4) a validated threshold for the inconclusive-image rule, more than one run of the image cross-check, and clinician confirmation of the reads it contested (a partial defence: it cannot catch a read both models get wrong); and (5) a repeat of the raw-image measurement with a held-back image set, since the public images may be in the model's training data.

# Chapter 7
# Conclusion

## 7.1 Conclusion

This project built and evaluated MedOrchestrate, a prototype that turns a mixed clinical input into a ranked, cited differential and decides, by a deterministic score, whether to refine its search or hand the case to a clinician. What was achieved and measured is an end-to-end system with 641 backend and 124 frontend tests; retrieval whose MRR@5 on 14 real agent queries rose from 0.685 to 0.863 through corrections found by measurement and fell to 0.764 when a larger knowledge base closed coverage gaps; the first measurement of BiomedCLIP's behaviour on 88 real films in this setting, with intervals, baselines and a held-out test whose gain was not reproduced on 26 fresh images; a confidence gate, found by a live run, that stops empty input from skipping review (all six abstention cases now go to a clinician, one before); and a second reading of the image that marks doubtful zero-shot reads as contested, though in the one end-to-end comparison it changed no review decision. The weakest results are also the most informative: zero-shot imaging is a coarse pre-filter, and a retrieval-based confidence needs an explicit check that the case contained something to reason from.

These conclusions are limited as Chapter 6 states. The system is a research prototype, not a validated clinical tool, and its contribution to Sustainable Development Goal 3 remains expected rather than demonstrated.

## 7.2 Future work

Table 7.1: Next steps, dependencies and proposed validation

| Next step | Depends on | Proposed validation |
|---|---|---|
| Repeat the full 31-case run at least three times with and without the image cross-check, and judge each; add a held-back image set for the raw-image comparison | Language-model quota; images no model has seen | Report the spread of confidence, review decisions and judge scores |
| Judge the 31 reports with a model of a different family, and spot-check by a clinician | A second provider key; clinician time | Agreement between the judges; check the cases the judge marked wrong |
| Collect at least 10 real clinician reviews; verify the answer key and image labels | Clinician time | Weight recalibration on real outcomes; inter-rater agreement |

| Next step | Depends on | Proposed validation |
|---|---|---|
| Calibrate the 0.65 threshold, weights and agreement signal | The repeats and clinician reviews | Reliability curve of confidence against review decisions |
| Widen the image benchmark (more atrophy films, any CT atrophy, a second normal head-CT subject, more normal films); have a clinician judge the retrieval relevance sets | More openly licensed images; clinician time | Same fresh-set design: plan fixed before scoring |
| Pin the container image's package versions and test it on a second machine; run the Claude path end to end | A second machine; a Claude key | Same cases through the container and each provider |

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# Appendix A
# Source Code and Reproducibility

The code, tests, documentation and the benchmark data are in the repository `https://github.com/hari18305/medorcha`, version `9d48c6e` at the time of writing. Secrets are not committed; an `.env.example` documents every setting.

## A.1 Setup and execution

```
python -m venv venv
venv\Scripts\activate
pip install -r requirements.txt
copy .env.example .env
  (set LLM_PROVIDER and the matching API key in .env)
python -m backend.rag.build_knowledge_base
python -m backend.rag.ingest
uvicorn backend.main:app --host 127.0.0.1 --port 8000
cd frontend && npm install && npm run dev
  (opens http://localhost:5173; or: docker compose up --build,
   then docker compose exec backend python -m backend.rag.ingest)

pytest -q backend
cd frontend && npm test
npm run test:e2e
python -m scripts.eval_retrieval
python -m scripts.build_imaging_benchmark
python -m backend.tests.imaging_benchmark
python scripts/run_cases_live.py OUT_DIR
```

The first group starts the system, the second runs the 641 backend and 124 frontend tests, the retrieval harness (offline) and the image benchmark; the last command needs a provider key.

## A.2 Supporting material

The imaging benchmark contains 88 real images (62 original and 26 added later): 6 under CC BY, 63 under CC BY-SA, 14 under CC0, 5 under Public domain. Each Commons image keeps its own licence, was downscaled to at most 768 px wide (the only change), and is credited with its author and licence in `data/benchmarks/imaging_real/ATTRIBUTION.md` (the six earlier images in `data/images/real/ATTRIBUTION.md`). The images are real people's scans, unidentified, and none of them validates the system.

# Appendix B
# Supporting Results

## B.1 Per-query retrieval results

Table B.1: Retrieval harness, final system: rank of the first relevant passage

| Query (case) | Dense rank | Final rank | Top two passages returned |
|---|---|---|---|
| CASE001 headache+HTN | 1 | 1 | hypertension overview, giant cell arteritis |
| CASE009 elderly warfarin fall | 16 | 3 | tia overview, subarachnoid hemorrhag |
| CASE010 child headache+URTI | 3 | 3 | secondary headache red, long covid headache |
| CASE011 anemia dizzy standing | 8 | 1 | anemia overview, fainting |
| CASE014 migraine | 1 | 1 | migraine overview, vestibular migraine di |
| CASE015 BPPV | 1 | 1 | bppv, menieres disease |
| CASE018 head injury+anticoag | 3 | 1 | chronic subdural hemat, subarachnoid hemorrhag |
| CASE020 cough smoker nodule | 1 | 5 | chest xray overview, pleural effusion chest |
| CASE022 pneumothorax | 3 | 1 | pneumothorax collapsed, pleural effusion chest |
| REAL003 headache+blurred vision | 2 | 1 | migraine overview, long covid headache |
| REAL004 memory+gait+atrophy | 1 | 3 | chronic subdural hemat, head ct overview |
| REAL005 chest pain, normal CXR | 1 | 2 | chest pain guideline 2, chest xray overview |
| REAL006 pneumothorax text+img | 3 | 1 | pneumothorax collapsed, pleural effusion chest |
| REAL007 pneumothorax img only | 1 | 1 | pneumothorax collapsed, atelectasis |

## B.2 Per-finding imaging results

Table B.2: Top-1 per finding on the original 62 images, before and after the caption change (counts of images)

| Modality | Finding | n | Top-1 before | Top-1 after | Top-3 after |
|---|---|---|---|---|---|
| CT | acute infarct | 3 | 2 | 1 | 3 |
| CT | hemorrhage | 5 | 0 | 3 | 4 |
| CT | mass lesion | 3 | 0 | 0 | 1 |
| CT | normal head CT, no acute findings | 3 | 3 | 1 | 3 |
| CT | sinus abnormality | 2 | 0 | 2 | 2 |
| CT | ventricular enlargement | 2 | 1 | 1 | 1 |
| MRI | acute infarct | 2 | 1 | 2 | 2 |
| MRI | hemorrhage | 2 | 0 | 1 | 1 |
| MRI | mass lesion | 4 | 0 | 4 | 4 |
| MRI | normal brain MRI, no acute findings | 2 | 1 | 0 | 2 |
| MRI | ventricular enlargement | 1 | 0 | 0 | 0 |
| MRI | white matter hyperintensity | 1 | 1 | 0 | 1 |
| CXR | atelectasis | 3 | 2 | 0 | 1 |
| CXR | cardiomegaly | 3 | 2 | 3 | 3 |
| CXR | consolidation | 3 | 0 | 0 | 3 |

| Modality | Finding | n | Top-1 before | Top-1 after | Top-3 after |
|---|---|---|---|---|---|
| CXR | hilar or mediastinal widening | 1 | 1 | 1 | 1 |
| CXR | hyperinflation consistent with emphysema | 2 | 0 | 2 | 2 |
| CXR | normal chest X-ray, no acute findings | 4 | 3 | 1 | 1 |
| CXR | pleural effusion | 3 | 2 | 3 | 3 |
| CXR | pneumothorax | 4 | 3 | 3 | 3 |
| CXR | pulmonary edema | 3 | 3 | 3 | 3 |
| CXR | pulmonary nodule or mass | 3 | 0 | 2 | 3 |
| CXR | rib fracture | 3 | 1 | 2 | 2 |

# Appendix C
# Worked Example

This appendix follows one case through the system, using the saved output of the 31-case live run (the version before the input-sufficiency gate). It illustrates the shape of the result; it does not measure quality.

## C.1 Input

Table C.1: Case CASE002 as submitted

| Field | Value |
|---|---|
| Patient | 52-year-old woman |
| Clinical text | recurrent headache and dizziness over the past month, history of hypertension |
| History (typed), laboratory values, report, image | hypertension; none; none; none |

## C.2 Output

The pipeline returned three diagnoses at confidence 0.757 (above the 0.65 threshold), with no conflicts, no optimizer rounds and live evidence present, and no review flag. Confidence is the case score of Section 5.2.8; the per-diagnosis figures are the model's own.

Table C.2: Ranked differential for CASE002

| Rank | Diagnosis | Per-diagnosis figure | Evidence cited |
|---|---|---|---|
| 1 | Hypertensive emergency / malignant hypertension | 0.65 | [RAG2] passage: severe headache, nausea, confusion and vision changes in malignant hypertension |
| 2 | Vestibular migraine | 0.55 | [RAG3] symptom passage on dizziness; [LIVE10] vestibular migraine abstract |
| 3 | Subarachnoid hemorrhage | 0.45 | [RAG1] subarachnoid bleeding is an emergency; high blood pressure is a risk factor |

Each citation tag becomes a link to its source in the interface. The first-ranked diagnosis rests on a general hypertension passage, not a guideline specific to this presentation, and the system leaves that visible.

# Appendix D
# Knowledge Base Inventory

The knowledge base holds 67 passages: 47 MedlinePlus pages (22 encyclopedia articles and 25 health topics) and 20 PubMed abstracts, retrieved on 12 and 13 November 2026 (the last 25 were added on the 13th, to cover conditions and imaging findings the first 42 did not). Each passage is stored with its URL, so every citation resolves to its source.

Table D.1: Knowledge-base passages

| # | Title | Source |
|---|---|---|
| 1 | Red and orange flags for secondary headaches in clinical practice: SNNOOP10... | PubMed PMID 30587518 |
| 2 | Clinical Practice Guideline: Benign Paroxysmal Positional Vertigo (Update) | PubMed PMID 28248609 |
| 3 | The 2020 International Society of Hypertension global hypertension practice... | PubMed PMID 32972800 |
| 4 | Orthostatic Hypotension in Adults With Hypertension: A Scientific Statement... | PubMed PMID 38205630 |
| 5 | Hypertensive Emergency | PubMed PMID 28372707 |
| 6 | Vestibular migraine: diagnostic criteria | PubMed PMID 23142830 |
| 7 | Guidelines for Management of Incidental Pulmonary Nodules Detected on CT Im... | PubMed PMID 28240562 |
| 8 | Guidelines for Management of Idiopathic Normal Pressure Hydrocephalus (Thir... | PubMed PMID 33455998 |
| 9 | Idiopathic intracranial hypertension: consensus guidelines on management | PubMed PMID 29903905 |
| 10 | Vascular contributions to cognitive impairment and dementia: a statement fo... | PubMed PMID 21778438 |
| 11 | 2021 AHA/ACC/ASE/CHEST/SAEM/SCCT/SCMR Guideline for the Evaluation and Diag... | PubMed PMID 34756653 |
| 12 | Tension headache | MedlinePlus encyclopedia |
| 13 | Migraine | MedlinePlus encyclopedia |
| 14 | Benign positional vertigo | MedlinePlus encyclopedia |
| 15 | Meniere disease | MedlinePlus encyclopedia |
| 16 | High blood pressure in adults - hypertension | MedlinePlus encyclopedia |
| 17 | Low blood pressure | MedlinePlus encyclopedia |
| 18 | Anemia | MedlinePlus encyclopedia |
| 19 | Transient ischemic attack | MedlinePlus encyclopedia |
| 20 | Subdural hematoma | MedlinePlus encyclopedia |
| 21 | Chronic obstructive pulmonary disease (COPD) | MedlinePlus encyclopedia |
| 22 | Pleural effusion | MedlinePlus encyclopedia |
| 23 | Head MRI | MedlinePlus encyclopedia |
| 24 | Head CT scan | MedlinePlus encyclopedia |
| 25 | Chest x-ray | MedlinePlus encyclopedia |
| 26 | Alzheimer disease | MedlinePlus encyclopedia |
| 27 | Dementia | MedlinePlus encyclopedia |
| 28 | Subarachnoid hemorrhage | MedlinePlus encyclopedia |
| 29 | Pulmonary tuberculosis | MedlinePlus encyclopedia |
| 30 | Headache | MedlinePlus health topic |
| 31 | Dizziness and Vertigo | MedlinePlus health topic |
| 32 | Ischemic Stroke | MedlinePlus health topic |

| # | Title | Source |
|---|---|---|
| 33 | Sinusitis | MedlinePlus health topic |
| 34 | Pneumonia | MedlinePlus health topic |
| 35 | Collapsed Lung | MedlinePlus health topic |
| 36 | Hydrocephalus | MedlinePlus health topic |
| 37 | Lung Cancer | MedlinePlus health topic |
| 38 | Hemorrhagic Stroke | MedlinePlus health topic |
| 39 | Traumatic Brain Injury | MedlinePlus health topic |
| 40 | Chest Injuries and Disorders | MedlinePlus health topic |
| 41 | Vestibular Neuritis: Recent Advances in Etiology, Diagnostic Evaluation, an... | PubMed PMID 30947184 |
| 42 | White matter hyperintensities associated with small vessel disease impair s... | PubMed PMID 28685621 |
| 43 | Is cardiomegaly on chest radiograph representative of true cardiomegaly: a... | PubMed PMID 29073657 |
| 44 | Pulmonary edema | MedlinePlus encyclopedia |
| 45 | Atelectasis | MedlinePlus encyclopedia |
| 46 | Cluster headache | MedlinePlus encyclopedia |
| 47 | Brain abscess | MedlinePlus encyclopedia |
| 48 | Heart Failure | MedlinePlus health topic |
| 49 | Brain Tumors | MedlinePlus health topic |
| 50 | Meningitis | MedlinePlus health topic |
| 51 | Chest Pain | MedlinePlus health topic |
| 52 | Fainting | MedlinePlus health topic |
| 53 | Fatigue | MedlinePlus health topic |
| 54 | Multiple Sclerosis | MedlinePlus health topic |
| 55 | Giant Cell Arteritis | MedlinePlus health topic |
| 56 | Concussion | MedlinePlus health topic |
| 57 | Brain Aneurysm | MedlinePlus health topic |
| 58 | Arteriovenous Malformations | MedlinePlus health topic |
| 59 | Aortic Aneurysm | MedlinePlus health topic |
| 60 | Long COVID | MedlinePlus health topic |
| 61 | Refractive Errors | MedlinePlus health topic |
| 62 | Pathophysiology and Diagnosis of Vertebrobasilar Insufficiency: A Review of... | PubMed PMID 28680502 |
| 63 | Long COVID headache | PubMed PMID 35915417 |
| 64 | Prevalence of musculoskeletal chest pain in the emergency department: a sys... | PubMed PMID 33838099 |
| 65 | Massive cerebral infarction | PubMed PMID 15860137 |
| 66 | Posterior reversible encephalopathy syndrome: evolving insights in diagnosi... | PubMed PMID 40818477 |
| 67 | Differential diagnosis of acute intracranial hemorrhage and calcification b... | PubMed PMID 37994049 |

# Appendix E
# API and Configuration Reference

## E.1 HTTP interface

Table E.1: Endpoints

| Method and path | Purpose |
|---|---|
| POST /diagnose | Run a case synchronously and return the final report |
| POST /diagnose?async=true | Accept a case, return 202 with a job identifier, run in the background |
| GET /diagnose/{id}/status | Step-by-step progress events and queue position for a running job |
| GET /diagnose/{id} | Return the stored report for a finished case |
| POST /review/{id} | Record a clinician decision (accept, override, annotate) and rating |
| GET /review/{id} | Return the stored review |
| GET /health | Liveness check |

## E.2 Configuration

Table E.2: Settings read from the environment (documented in .env.example)

| Setting | Default | Meaning |
|---|---|---|
| LLM_PROVIDER | gemini | Language-model provider: gemini, anthropic, bedrock, vertex or mistral |
| GEMINI_MODEL_ID | (set in .env) | Model used for every live run in this report: gemini-3.5-flash-lite |
| GEMINI_RPM_BUDGET | 12 | Client-side request budget per minute |
| RERANK_MODEL | ncbi/MedCPT-Cross-Encoder | Cross-encoder used to rerank the dense top twenty |
| RERANK_ORDER | ce | ce: cross-encoder order; rrf: reciprocal rank fusion with the dense rank |
| IMAGING_FINDING_PROMPTS | showing | Descriptive finding captions; label restores the bare labels (used for the demonstration) |
| IMAGING_CROSS_CHECK | off | on: the multimodal model's own reading of the image can mark a zero-shot read contested |
| IMAGING_MIN_TOP_SCORE | 0.40 | A zero-shot read whose top label scores below this is inconclusive |
| FINDING_RELATIVE_THRESHOLD | 0.6 | Secondary findings must reach this share of the top score |
| CASE_MEMORY_MIN_SCORE | 0.6 | BioBERT similarity floor for a past case to be retrieved |
| VISUAL_MEMORY_MIN_SCORE | 0.85 | BiomedCLIP similarity floor for an image precedent (same modality) |
| EVIDENCE_LLM_SUMMARIES | false | Summarise live evidence with the language model |
| DATABASE_URL | sqlite:///./medorchestrate.db | Case, review and weight storage |
| QDRANT_LOCAL_PATH / QDRANT_URL | embedded local store | Vector store location |

# Appendix F
# Test Inventory and Live-Case Results

## F.1 Automated tests by module

Table F.1: Backend tests collected by pytest

| Module | Covers | Tests |
|---|---|---|
| test_answer_key | answer key integrity | 5 |
| test_api | HTTP endpoints, async jobs, validation | 24 |
| test_case_memory | text and visual case memory | 18 |
| test_clinical_agent | clinical extraction, history and lab merge | 12 |
| test_confidence | formula, gate, threshold, weights | 54 |
| test_embeddings | BioBERT embedding wrapper | 2 |
| test_errors | | 17 |
| test_evidence_agent | four live sources, timeouts, retries, deduplication | 48 |
| test_fixture_cases | the 27 case fixtures run through the pipeline | 48 |
| test_fusion_agent | citations, no invented sources, image attachment | 26 |
| test_imaging_agent | modality and finding logic, unsupported modality | 42 |
| test_imaging_benchmark | benchmark scoring, intervals, baselines | 25 |
| test_imaging_prompt_experiment | split, selection rule, one-time test guard | 23 |
| test_inference | | 4 |
| test_ingest | knowledge-base ingest | 17 |
| test_integration | end-to-end flows with faked services | 28 |
| test_jobs | job registry, queue, pruning | 15 |
| test_llm_client | four provider clients, retries, errors | 36 |
| test_llm_judge | judge metric parsing | 26 |
| test_logging | tracing and redaction | 14 |
| test_optimizer_agent | query refinement rounds and gate | 16 |
| test_orchestrator | step order, skips, progress events, failures | 37 |
| test_rag_agent | dense retrieval, rerank, calibration | 16 |
| test_reranker | reranker models and ordering | 14 |
| test_retrieval_harness | | 5 |
| test_review_flow | accept, override, annotate, rating | 15 |
| test_run_evaluation | evaluation runner | 14 |
| test_storage | SQLite persistence | 27 |
| test_weight_tuner | recalibration rules | 13 |
| **Total** | | **641** |

# F.2 The 31 live cases

Table F.2: Per-case results: first run (before the input-sufficiency gate) and the run on the final system

| Case | Input | First run | Final run | Review | Rounds | Live ev. |
|---|---|---|---|---|---|---|
| CASE001 | text only | 0.76 | 0.75 | no | 0 | yes |
| CASE001 with image | text and image | 0.66 | 0.65 | no | 3 | yes |
| CASE002 text only | text only | 0.76 | 0.68 | no | 0 | yes |
| CASE003 image only | image only | 0.67 | 0.50 | yes | 0 | yes |
| CASE004 text and image | text, image | 0.69 | 0.73 | no | 0 | yes |
| CASE005 text and medical report | text, report | 0.83 | 0.75 | no | 0 | yes |
| CASE006 medical report only | report only | 0.69 | 0.69 | no | 1 | yes |
| CASE007 high confidence candidate | text | invalid | 0.82 | no | 1 | yes |
| CASE008 low confidence candidate | text | 0.68 | 0.60 | yes | 0 | yes |
| CASE009 elderly patient | text | 0.69 | 0.64 | yes | 3 | yes |
| CASE010 pediatric patient | text | 0.62 | 0.74 | no | 2 | yes |
| CASE011 with labs | text | 0.74 | 0.66 | no | 0 | yes |
| CASE012 extensive history | text | 0.84 | 0.68 | no | 0 | yes |
| CASE013 minimal no content | empty input | 0.79 | 0.00 | yes | 0 | no |
| CASE014 headache focused | text | 0.74 | 0.78 | no | 0 | yes |
| CASE015 dizziness focused | text | 0.78 | 0.75 | no | 0 | yes |
| CASE016 multiple symptoms with image | text, image | 0.67 | 0.61 | yes | 3 | yes |
| CASE017 ct image only | CT, image only | 0.69 | 0.50 | yes | 0 | yes |
| CASE018 ct text and image | text, image | 0.71 | 0.70 | no | 0 | yes |
| CASE019 ct text report and image | text, report, image | 0.70 | 0.65 | no | 0 | yes |
| CASE020 chest xray text and image | text, image | 0.68 | 0.61 | yes | 3 | yes |
| CASE021 chest xray image only | chest X-ray, image only | 0.57 | 0.50 | yes | 0 | yes |
| CASE022 chest xray text report and image | text, report, image | 0.76 | 0.67 | no | 0 | yes |
| CASE023 unsupported modality image | image of an unsupported modality | 0.77 | 0.75 | no | 0 | yes |
| REAL001 ct head text and image | text, image | 0.71 | 0.65 | no | 0 | yes |
| REAL002 ct head image only | CT, image only | 0.69 | 0.50 | yes | 0 | yes |
| REAL003 ct head text report and image | text, report, image | 0.77 | 0.61 | yes | 3 | yes |
| REAL004 brain mri text and image | text, image | 0.65 | 0.60 | yes | 3 | yes |
| REAL005 chest xray normal text and image | text, image | 0.70 | 0.60 | yes | 3 | yes |
| REAL006 chest xray pneumothorax text and | text, image | 0.77 | 0.73 | no | 0 | yes |
| REAL007 chest xray pneumothorax image on | chest X-ray, image only | 0.68 | 0.50 | yes | 0 | yes |

Input labels are inferred from the case names. CASE013 (empty) shows the gate defect described in Section 6.3.6: confidence 0.79 with no diagnosis in the first run, 0.00 with review in the final run. CASE007 was invalid in the first run because the provider returned quota errors; it is excluded from the first run's means.